$\bar{\nu}_e$ Appearance Analysis with the 3-Flavour Joint Fit Framework for T2K Run 1-9c (1.4938×10^{21}-POT Neutrino and 1.1236×10^{21}-POT Anti-Neutrino) and Run 1-9d (1.4938×10^{21}-POT Neutrino and 1.6346×10^{21}-POT Anti-Neutrino) Data Sets

Costas Andreopoulos[a,b], Christopher Barry[b], Francis Bench[b], Andrew Chappell [d], Thomas Dealtry[e], Steve Dennis[a,b], Davide Sgalaberna[f], Raj Shah[a,c]

[a] *Science & Technology Facilities Council (STFC), Rutherford Appleton Laboratory, Harwell Oxford Campus, OX11 0QX, UK*
[b] *University of Liverpool, Physics Department, Liverpool, L69 7ZE, UK*
[c] *University of Oxford, Subdepartment of Particle Physics, Oxford, OX1 3RH, UK*
[d] *University of Warwick, Physics Department, Coventry, CV4 7AL, UK*
[e] *Lancaster University, Physics Department, Lancaster, LA1 4YB, UK*
[f] *University of Geneva, Departement de physique nucleaire et corpusculaire, 1211 Geneve, Switzerland*

Abstract

We report the results of the $\bar{\nu}_e$ appearance analysis using the 3-flavour, 5 sample $\nu/\bar{\nu}$ analysis framework, performed by the *VALOR* group, on the combined Run 1-9c data set, using data taken in both neutrino and antineutrino beam mode. The goal of this analysis is to search for $\bar{\nu}_e$ appearance in a $\bar{\nu}_\mu$ beam and is an update to [1] The joint fit framework is used, whose details can be found in [2]. A joint analysis of all the Super-K samples, in both neutrino and antineutrino beam modes, is used in order to constrain the oscillation and systematic parameters, achieving a better sensitivity to $\bar{\nu}_e$ appearance. The Run 1-9c data set corresponds to an integrated J-PARC neutrino beam exposure of 1.4938×10^{21} POT in neutrino mode and 1.1236×10^{21} POT in antineutrino mode.

A parameter β, that multiplies the $\bar{\nu}_\mu \to \bar{\nu}_e$ oscillation probability by either 0 or 1, is introduced in order to characterise the significance to $\bar{\nu}_e$ appearance. All the oscillation parameters, mass hierarchy included, are marginalised. The significance of the result is shown as p-values and a Bayes factor. Both a rate+shape and a rate-only analysis are performed.

Run 1-9c Analysis

Using the Asimov A parameters (see Tab 2), this analysis predicts 268.5 ± 13.2 (syst) for neutrino μ-like, 95.5 ± 4.1 (syst) for antineutrino μ-like, 73.8 ± 7.1 (syst) for neutrino e-like, 11.8 ± 0.9 (syst) for antineutrino e-like and 6.9 ± 1.3 (syst) for neutrino ν_eCC1π^+-like events in Super K and observes respectively 243 for neutrino μ-like, 102 for antineutrino μ-like, 75 for neutrino e-like, 9 for antineutrino e-like and 15 for neutrino ν_eCC1π^+-like events.

For the Run 1-9c set and taking no $\bar{\nu}_e$ appearance as our null hypothesis we find a p-value of

0.209 ± 0.003 for the rate-only analysis and a p-value of 0.215 ± 0.003 for the rate+shape analysis. Taking $\bar{\nu}_e$ appearance according to the best knowledge of the PMNS matrix as our null hypothesis we find a p-value of 0.210 ± 0.003 for the rate-only analysis and a p-value of 0.092 ± 0.003 for the rate+shape analysis. For both hypotheses, the uncertainties in the oscillation and systematic parameters are accounted for by generating a large number of pseudo-data sets where their values are randomly varied according to their prior probability distributions. The Bayes factor, given by the rate+shape analysis, was found to be 2.12 indicating a poor preference for the no $\bar{\nu}_e$ appearance hypothesis over the PMNS $\bar{\nu}_e$ appearance hypothesis.

Full Run 1-9 Analysis

An updated analysis was performed using full Run 1-9 statistics (an increase of 46 % antineutrino exposure). Using the Asimov A parameters, the updated predicts 272.4 ± 13.9 (syst) for neutrino μ-like, 139.5 ± 6.1 (syst) for antineutrino μ-like, 72.8 ± 6.8 (syst) for neutrino e-like, 16.8 ± 1.3 (syst) for antineutrino e-like and 6.9 ± 1.3 (syst) for neutrino ν_eCC1π^+-like events in Super K and observes respectively 243 for neutrino μ-like, 140 for antineutrino μ-like, 75 for neutrino e-like, 15 for antineutrino e-like and 15 for neutrino ν_eCC1π^+-like events.

For the full Run 1-9 data set and taking no $\bar{\nu}_e$ appearance as our null hypothesis we find a p-value of 0.059 ± 0.002 for the rate-only analysis and a p-value of 0.0163 ± 0.0009 for the rate+shape analysis. Taking $\bar{\nu}_e$ appearance according to the best knowledge of the PMNS matrix as our null hypothesis we find a p-value of 0.321 ± 0.003 for the rate-only analysis and a p-value of 0.300 ± 0.004 for the rate+shape analysis. For both hypotheses, the uncertainties in the oscillation and systematic parameters are accounted for by generating a large number of pseudo-data sets where their values are randomly varied according to their prior probability distributions. The Bayes factor, given by the rate+shape analysis, was found to be 0.149 indicating a substantial preference for the PMNS $\bar{\nu}_e$ appearance hypothesis over the no $\bar{\nu}_e$ appearance hypothesis.

Updates to TN from Run 1-9c (TN-356 v4) to Full Run 1-9 (TN-356 v5)

- Updated the description of the marginal likelihood calculation to include a description of the separate treatment of the RHC1Re sample and the control samples used in both Run 1-9c and Run 1-9d analyses
- Split up the section describing the calculation of marginal likelihood (previously sec 1.3) into calculation of likelihood (sec 1.3) and weighted marginal likelihood (sec 2.2.1)
- Added a section on the full Run 1-9 update (sec 6)
- Updated the optimised priors used to generate fake data distribution toys (appendix A.1)
- Updated the p-value stability test (appendix B.1)
- Added a comparison of the different toy generation methods used by VALOR and P-theta (appendix C)
- Added an appendix comparing the sensitivity and data results obtained using full Run 1-9 statistics against those obtained using only Run 1-9c statistics (appendix D)

Contents

1 Introduction **6**
- 1.1 Improvements to the Previous Analysis . . . 6
- 1.2 Construction of Nominal Super-K Monte Carlo Templates . . . 7
- 1.3 Calculation of the Likelihood . . . 8

2 $\bar{\nu}_e$ Appearance Methodology **8**
- 2.1 Rate-only Analysis Method . . . 9
- 2.2 Rate+shape Analysis Method . . . 10
  - 2.2.1 Calculation of the Marginal Likelihood . . . 10
  - 2.2.2 Calculation of the Test Statistic and p-value . . . 11
- 2.3 Differences Between the Rate-only and Rate+shape Analysis Methods . . . 11
- 2.4 Comparison to the P-theta Analysis . . . 12
- 2.5 Statistical Uncertainty on the p-values . . . 12
- 2.6 Bayes Factor . . . 13

3 Sensitivity to $\bar{\nu}_e$ Appearance **14**

4 Results of the $\bar{\nu}_e$ Appearance Analysis **17**

5 Summary **22**

6 Updated Analysis Using the Full T2K Run 1-9 Dataset **23**
- 6.1 Sensitivity to $\bar{\nu}_e$ Appearance . . . 23
- 6.2 Results of the $\bar{\nu}_e$ Appearance Analysis . . . 27

Appendices **31**

A Optimisation of Oscillation Parameter Priors **31**
- A.1 Updated Priors for the Full Run 1-9 Analysis . . . 32

B p-value Stability **32**
- B.1 Improved p-value Stability Study using the Full Run 1-9 Sensitivity . . . 33

C Toy Generation Method Comparison **34**

D Comparison of Full Run 1-9 vs. Run 1-9c Results **37**
- D.1 Comparison of Expected Sensitivities . . . 37
- D.2 Comparison of Data Results . . . 37

E Comparison Between VALOR and P-theta Results using Full Run 1-9 Statistics 41
E.1 Comparison of Expected Sensitivities 41
E.2 Comparison of Data Results 41

Parameter(s)	Prior	Range for marginalisation toys	Range for fake-data toys
$\sin^2\theta_{23}$	Uniform	$[0.3; 0.7]$	$[0.42; 0.62]$
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	Gaussian	0.0857 ± 0.0046	0.0857 ± 0.0046
$\sin^2 2\theta_{12}$	Gaussian	0.846 ± 0.021	0.846 ± 0.021
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{13}\rvert$ (IH)	Uniform	$[2.3; 2.7] \times 10^{-3}$ eV^2/c^4	$[2.30, 2.65] \times 10^{-3}$ eV^2/c^4
Δm^2_{21}	Gaussian	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4
δ_{CP}	Uniform	$[-\pi; +\pi]$	$[-\pi, 1.1]\&[1.8, \pi]$
Mass Hierarchy	Bernoulli	0.5 normal, 0.5 inverted	0.5 normal, 0.5 inverted

Table 1: Treatment of the oscillation parameters in the $\bar{\nu}_e$ appearance analysis. All the gaussian priors are from [5]. The parameter $\sin^2 2\theta_{13}$ is constrained with the measurement of reactor experiments using the Gaussian prior [5]. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [5], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [6]. When generating fake-data toys used to fill the test-statistic distributions, the parameters that are thrown according to flat prior probability distributions have a restricted range compared to the priors used when generating marginalisation toys. This is purely to decrease the required CPU time when generating fake-data toys whilst having minimal impact on the p-values produced. This is detailed in appendix A.

1. Introduction

In this technical note, we report the results of the $\bar{\nu}_e$ appearance analysis using the 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the VALOR group, on the combined Run 1-9c data set, using data taken in both neutrino mode (1.4938×10^{21}), with the polarity set to forward horn current (FHC), and antineutrino (1.1236×10^{21}) mode with polarity set to reverse horn current (RHC). The same procedure used in the 2016 $\bar{\nu}_e$ appearance analysis [1] is used, whilst working in the analysis framework outlined in [2]. A parameter β, that multiplies by either 0 or 1 the $\bar{\nu}_\mu \to \bar{\nu}_e$ oscillation probability, is used in order to obtain the significance to $\bar{\nu}_e$ appearance. The priors used to marginalise the oscillation parameters and generate fake-data toys can be found in Tab 1.

The significance to $\bar{\nu}_e$ appearance is given as a p-value. Two different test statistics are used. The total number of observed events (N_{obs}) in the antineutrino mode e-like sample, without using any information from the reconstructed energy shape, is used to calculate the rate-only p-value. $\Delta\chi^2 = \chi^2(\beta = 0) - \chi^2(\beta = 1)$ is used calculate the rate+shape p-value, where both rate and shape information from the RHC mode single-ring e-like sample and a constraint on the oscillation parameters from the other four samples (single-ring μ-like and single-ring e-like, both in FHC and RHC modes and FHC $\nu_e CC1\pi$-like) are used. Where the χ^2 values are calculated by marginalising over all systematic and oscillation parameters, including the mass hierarchy (see section 2.2.1)

1.1. Improvements to the Previous Analysis

There have been a number of improvements since this analysis was previously performed, in 2016 [1]:

- The FHC exposure has increased by 113 % from 7.002×10^{20} to 1.4938×10^{21}
- The RHC exposure has increased by 50 % from 7.471×10^{20} to 1.1236×10^{21}
- An additional Super-K event sample has been added, neutrino mode ν_eCC1π-like

- The Super-K event reconstruction algorithm has changed from using APFit to FitQun [7]
- New cross-section model parameters are implemented, as detailed in [2]

1.2. Construction of Nominal Super-K Monte Carlo Templates

The Monte Carlo templates (predicted event spectra for each reaction mode produced by Monte Carlo simulation and binned in kinematic variables) used in this analysis were constructed in an analogous way to the procedure used for the 3-flavour joint analysis in [2]. The parameter of interest β which reweights the $\bar{\nu}_e$ appearance signal component is kept fixed to either 0 (corresponding to no appearance) or 1 (corresponding to appearance consistent with the current best knowledge of the PMNS matrix given by the simultaneous T2K+reactor Run 1-8 joint fit result) and is applied to all the Super-K samples, though its impact is negligible in all the samples except the antineutrino mode e-like sample.

In this analysis a comparison of the reconstructed energy (E_r) and reconstructed lepton angle (θ) spectrum of single ring events, including the ν_e CC1π^+-like sample, observed at Super-K with the predicted spectrum is used to build the test statistic to calculate the significance to $\bar{\nu}_e$ appearance. For μ-like samples the templates are built as a function of the reconstructed neutrino energy, as in previous analyses, and a single θ bin is used. The predicted number $N_{SK;r,\theta}$ of RHC single-ring e-like events in the r-th and θ-th bin is computed for each selected sample as follows:

$$N_{SK;r,\theta} = \sum_{m} \sum_{t} \sum_{r\prime} \left[\delta\left(m - m_{\bar{\nu}_\mu \to \bar{\nu}_e}\right)(\beta - 1) + 1\right] \cdot P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime,\theta;\vec{f}} \cdot N^{MC}_{SK;m;r\prime,\theta;t} \quad (1)$$

In equation 1, $N^{MC}_{SK;m;r\prime,\theta;t}$ is the input Super-K Monte Carlo (MC) template containing the number of events in the single ring MC sample with true reaction mode m in the true energy bin t, the reconstructed energy bin $r\prime$ and reconstructed lepton angle bin θ. $\delta\left(m - m_{\bar{\nu}_\mu \to \bar{\nu}_e}\right)$ is the Dirac delta function, equal to one when the reaction mode corresponds to the $\bar{\nu}_\mu \to \bar{\nu}_e$ oscillation and zero otherwise. $S_{m;t;r\prime,\theta;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on a vector of nuisance (systematic) parameters $\vec{f}$, which is function of the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$, and the reconstructed lepton angle bin θ. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the Super-K reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the Super-K MC template which corresponds to mode m; This is exactly how the analysis templates are built in the joint fit analysis, with the only difference that the $\bar{\nu}_\mu \to \bar{\nu}_e$ component can be switched on or off by the parameter β in each Super-K sample. In Fig 3 the predicted spectra are compared with the observed data. Also the 2-dimensional spectrum of the $\bar{\nu}_\mu \to \bar{\nu}_e$ component, that can be switched on or off by the parameter β is shown.

1.3. Calculation of the Likelihood

In the analyses presented here both the likelihood and the marginal likelihood are used. The marginal likelihood calculation uses details of the $\bar{\nu}_e$ appearance methodology so is explained separately, see section 2.2.1.

First the log-likelihood function is calculated for each sample, s as follows:

$$-2\ln\lambda_s\left(Data;Toy\right) = 2\cdot\sum_{i=0}^{N-1}\left(n_{s,i}^{obs}\cdot\ln\left(n_{s,i}^{obs}/n_{s,i}^{exp}\right)+\left(n_{s,i}^{exp}-n_{s,i}^{obs}\right)\right) \quad (2)$$

where $n_{s,i}^{obs}$ is the observed number of data events in the ith bin in reconstructed energy and lepton angle of sample s and $n_{s,i}^{exp} = n_{s,i}^{exp}(\beta;\mathbf{a})$ is the corresponding expected number of events in a toy, N is the number of reconstructed energy and lepton angle bins and $\mathbf{a}$ is a vector of systematic parameters. A 'toy' is a simulated pseudo-data set containing a particular set of β and $\mathbf{a}$ values and with non-integer n^{exp} generated from these values.

The total likelihood is then given by the sum of $-2\ln\lambda_s\left(Data;Toy\right)$ from all of the Super-K samples

$$-2\ln\lambda\left(Data;Toy\right) = \sum_{s=1}^{N_{samples}} -2\ln\lambda_s\left(Data;Toy\right) \quad (3)$$

where $N_{samples}$ corresponds to the number of Super-K samples used in the analysis.

2. $\bar{\nu}_e$ Appearance Methodology

The significance of $\bar{\nu}_e$ appearance is given as a p-value by introducing the parameter β in order to compare the data to two null hypotheses: a) No $\bar{\nu}_e$ appearance ($\beta = 0$) and b) $\bar{\nu}_e$ appearance according to the current best knowledge of the PMNS matrix ($\beta = 1$), where both hypotheses fully account for uncertainties in the values of the oscillation parameters. Two different test statistics are used: rate-only and rate+shape.

An ensemble of toys are generated for each null hypothesis, randomising nuisance parameters according to the priors in Tab 1 and applying statistical fluctuations. In order to constrain the oscillation and systematic model parameters using T2K data from the "control samples" (neutrino mode single-ring and multi-ring e-like and both neutrino and antineutrino mode single-ring μ-like), the toy distributions are filled according to the following method, which ensures that the test-statistic distributions are filled with only the most likely toys and maintains correlations between all nuisance parameters in the 4 "control samples":

1. A large number ($> 1M$) of toy experiments are produced taking random parameters from the distributions in Tab 1
2. The 4 sample (i.e. excluding antineutrino mode single-ring e-like) likelihood of the data given the toy is calculated for each toy using equation 4. Then the maximum likelihood, λ_{max}, is stored.

3. A proposed toy experiment, $T_{proposed}$, is produced taking random parameters from the distributions in Tab 1.
4. The 4 sample likelihood of the data given $T_{proposed}$ is found: $\lambda_{proposed} = \lambda_{control}(Data; Toy)$.
5. A random number R is drawn uniformly from $\{0 : 1\}$.
6. If $\frac{\lambda_{proposed}}{\lambda_{max}}$ is greater than R, $T_{proposed}$ is accepted and used in the distribution.
7. The procedure is repeated from step 3 until 20k of $T_{proposed}$ toys have been accepted, denoted $T_{accepted}$. The resulting set of $T_{accepted}$ toys are constrained by the likelihood of the "control sample" data given the toy and only contain the most likely toys. Toys generated by this method will be referred to as data-constrained toys.
8. The rate-only and rate+shape test-statistics are calculated according to sections 2.1 and 2.2 for each $T_{accepted}$ to produce test-statistic distributions.

$$\lambda_{control}\left(Data; Toy\right) = \prod_{s \neq RHC1Re} \lambda_s\left(Data; Toy\right) \tag{4}$$

It is worth noting that this method of filling the toy distributions is statistically equivalent to performing a likelihood ratio fit of the "control samples" and using the posterior distribution of the nuisance parameters as a prior to extract the systematic parameters for a toy experiment.

At this point, the rate-only and rate+shape analysis methods diverge.

2.1. Rate-only Analysis Method

The total number of observed events, N_{obs}, in the antineutrino mode e-like sample, without using any information from the reconstructed energy and angle distribution, is used to quote the rate-only p-value.

The expected distribution of antineutrino mode single-ring e-like events, under statistical and systematic fluctuations, is produced using 20k (throws of nuisance parameters) x 10k (throws of statistical fluctuations) toy data sets to ensure the nuisance parameter space was scanned effectively. In this method, the toy experiments used for the 20k nuisance parameter throws are the sampled toys described in section 2. For each of these toys, 10k statistical throws are performed using a Poisson distribution with mean equal to the number of antineutrino mode single-ring e-like events in the toy. The result of each statistical throw, N_{exp}, is stored in a rate-only test-statistic distribution for each toy, $f_{itoy}\left(N_{exp}\right)$.

After performing statistical throws, a p-value is calculated for each $f_{itoy}\left(N_{exp}\right)$. For the $\beta = 0$ null hypothesis, a right-tailed p-value is used (see equation 5), while for the $\beta = 1$ null hypothesis, a left-tailed p-value is used (see equation 6). For each null hypothesis, the mean of the rate-only p-values (averaged over each toy) is quoted as the result. This is equivalent to storing all statistical throws for all toys in a single distribution and calculating a single p-value from this, but is done to increase computational efficiency.

A right-tailed p-value is used for the $\beta = 0$ hypothesis as the $\beta = 0$ test-statistic distribution is composed solely of background events. The p-value is the probability of obtaining a result as

extreme or more extreme than the observed result under the null hypothesis, so using a right-tailed p-value gives the probability of observing a fluctuation under the $\beta = 0$ hypothesis in the background with equal to or more events than observed in the antineutrino mode single-ring e-like sample. An upwards fluctuation such as this would be less consistent with coming from the expected number of events from a sample containing only background events. Therefore, if this p-value is lower than some threshold (usually set to $5\,\sigma$), it could be claimed that the observation is incompatible with having come from the null hypothesis and the hypothesis is rejected.

Similarly, a left-tailed p-value is used for the $\beta = 1$ hypothesis as the $\beta = 1$ test-statistic distribution is composed of both background and signal events, so the left-tailed p-value gives the probability of observing a fluctuation under the $\beta = 1$ hypothesis with equal to or fewer events than were observed in the antineutrino mode single-ring e-like sample. A downwards fluctuation such as this would be less consistent with coming from the expected number of events from a sample containing both background and signal events.

$$\text{p-value(right tail)} = \frac{\int_{N_{obs}}^{+\infty} f\left(N_{exp}\right) dN_{exp}}{\int_{-\infty}^{+\infty} f\left(N_{exp}\right) dN_{exp}} \tag{5}$$

$$\text{p-value(left tail)} = \frac{\int_{-\infty}^{N_{obs}} f\left(N_{exp}\right) dN_{exp}}{\int_{-\infty}^{+\infty} f\left(N_{exp}\right) dN_{exp}} \tag{6}$$

As the distribution of the statistic is populated with such a large sample, the statistical uncertainty on the p-value is very small, of the order of 10^{-3}.

2.2. Rate+shape Analysis Method

In calculating the rate+shape test statistic, the marginal likelihood is used and is calculated according to the following method.

2.2.1. Calculation of the Marginal Likelihood

The marginal likelihood is computed by considering an ensemble of MC predictions, generated based on priors on systematic parameters, as in [2], and priors on oscillation parameters from Tab 1. Together, the oscillation and systematic parameters, including the mass hierarchy, are considered to be nuisance parameters. The dependence of the likelihood on the nuisance parameters is eliminated during the marginalisation process so that the likelihood is only dependant on β. The marginal likelihood is then calculated as the weighted mean of the antineutrino mode single-ring e-like sample likelihood evaluated at N_{marg} throws of all nuisance parameters, $\mathbf{a}$:

$$\lambda_{marg}\left(\beta\right) = \frac{1}{N_{marg}} \sum_{j=1}^{N_{marg}} \lambda_{RHC1Re}(T_{accepted}; \mathbf{a}_j, \beta) \times \lambda_{control}(Data; \mathbf{a}_j, \beta) \tag{7}$$

Where $\lambda_{RHC1Re}(T_{accepted}; \mathbf{a}_j, \beta)$ denotes the likelihood of each $T_{accepted}$ in the antineutrino mode single-ring e-like sample for the jth set of nuisance parameter throws, $\mathbf{a}_j$. This is weighted by the control sample likelihood of the data given the jth set of nuisance throws, $\lambda_{control}(Data; \mathbf{a}_j, \beta)$.

Weighting the likelihood in this way is equivalent to using the control sample data as a prior constraint when throwing the nuisance parameters.

In the limit of large statistics, $-2\ln\lambda$ asymptotically converges to a χ^2 distribution [12]. From this point onwards, the marginal log-likelihood will be instead written as χ^2_{marg}.

$$\chi^2_{marg}(\beta) = -2\ln\lambda_{marg}(\beta) \tag{8}$$

In this analysis, 200k nuisance parameter throws are used to determine the marginal likelihood in order to explore the whole space of the systematic parameters. More throws are used compared to the joint fit analysis, because in this analysis the mass hierarchy, δ_{CP} and all other oscillation parameters are marginalised at the same time whereas they were not in the joint analysis. Furthermore these parameters have a large impact on the $\nu_\mu \to \nu_e$ component which is a significant background in the null hypotheses.

2.2.2. Calculation of the Test Statistic and p-value

The rate+shape analysis method uses log-likelihood ratio between the $\beta = 0$ and $\beta = 1$ $\bar{\nu}_e$ appearance null hypotheses as the test-statistic:

$$\Delta\chi^2 = \chi^2_{marg}(\beta = 0) - \chi^2_{marg}(\beta = 1) \tag{9}$$

This test statistic simply compares the oscillation with the non-oscillation hypothesis but, unlike the $\Delta\chi^2$ values given in the 3-flavour joint analysis [2], this does not use the best-fit χ^2 value, as β is not fit for specifically. For this reason the distribution of $\Delta\chi^2$ is expected to take both negative and positive values, where $\Delta\chi^2 < 0$ (> 0) corresponds to the spectra being more consistent with the $\beta = 0$ (1) hypothesis than $\beta = 1$ (0). It is worth noting that the parameter β is not fit, but is fixed either to 0 or 1.

The $\Delta\chi^2$ distribution is produced by first generating 20k toys produced using the method described in section 2. The toys are then marginalised using 200k marginalisation toys generated from the priors in Tab 1 and $\chi^2_{marg}(\beta)$ is calculated according to equation 8. This incorporates the uncertainty on nuisance parameters on the measurement and distributions of the test-statistic.

Once a $\Delta\chi^2$ value is calculated for each toy, it is stored in the test-statistic distribution, $f\left(\Delta\chi^2\right)$, which is integrated to obtain p-values, for example the right-tailed p-value for the $\beta = 0$ null hypothesis shown in equation 10.

$$\text{p-value(right tail)} = \frac{\int_{\Delta\chi^2_{obs}}^{+\infty} f\left(\Delta\chi^2\right) d\Delta\chi^2}{\int_{-\infty}^{+\infty} f\left(\Delta\chi^2\right) d\Delta\chi^2} \tag{10}$$

2.3. Differences Between the Rate-only and Rate+shape Analysis Methods

The number of statistical throws from each toy data set is different between the rate-only and rate+shape analyses. The rate+shape analysis uses no single statistical fluctuations when generating each fake-data toy.

It is worth pointing out that sometimes the rate-only p-value can be smaller than the rate+shape. This can be due to both the discreteness of the rate-only p-value or to the shape information (which the rate-only analysis does not have access to) which can discredit the $\bar{\nu}_e$ appearance hypothesis. For this reason it is important to quote both the p-values as our sensitivity.

2.4. Comparison to the P-theta Analysis

Both the rate-only and rate+shape analyses were validated against what was found by the P-theta group. In the Run 1-9c analysis for the rate-only analysis we see good agreement between the two analyses, as expected. For the rate+shape analysis, the difference between P-theta and VALOR is mainly due to the e-like samples. For VALOR the e-like samples are binned in $E_{reco} - \theta$ analysis templates whereas for P-theta they are binned in $p - \theta$. When the P-theta group runs the analysis using $E_{reco} - \theta$, the results of VALOR and P-theta agree reasonably well (see Ref [15] for the Run 1-9c analysis). For the full Run 1-9 analysis, good agreement between VALOR and P-theta is found, as shown in appendix E.

In the Run 1-9c analysis, the remaining differences were thought to be due to the different toy generation methods used by VALOR and P-theta: Instead of sampling the toy parameter space around the maximum control sample likelihood to accept/reject toys, the P-theta analysis accepts all toys and applies the control sample likelihood as a weight when calculating p-values. This affect is studied in appendix C using full Run 1-9 statistics.

In addition, the P-theta group performed fake-data studies to assess the impact of uncertainties in the neutrino-nucleus interaction model and found no evidence of any significant bias [8].

2.5. Statistical Uncertainty on the p-values

The statistical uncertainty on the p-value is calculated assuming that the number of entries in each bin of the expected distribution of the test statistic follow a Poisson distribution. The p-value uncertainty is then derived from the statistical uncertainty of the population. As the p-value is calculated from the fraction of trials which fall to the right of the data statistic, the p-value is expected to be binomially distributed. Given a sufficiently large number of trials (N), the p-value is expected to take a gaussian distribution with a mean of the p-value and a standard deviation of $\sqrt{p(1-p)/N}$.

The uncertainty is calculated from the first order Taylor series expansion assuming no correlation between bins in the test-statistic distribution and the number of events in each bin follows a Poisson distribution. The calculation of the uncertainty summing over all toy experiments, for the right sided rate-only p-value, is shown as an example and is as follows:

The p-value calculation shown in equation 5 is written in terms of an integral over the test-statistic distribution. This is re-written as a sum of all bins that contribute to the right-tail of the distribution

$$\text{p-value(right tail)} = p = \frac{\int_{N_{obs}}^{+\infty} f\left(N_{exp}\right) dN_{exp}}{\int_{-\infty}^{+\infty} f\left(N_{exp}\right) dN_{exp}} = \frac{\sum_{i=0}^{n(right-tail)} n_i}{\sum_{i=0}^{n(total)} n_i} \tag{11}$$

Where n_i is the the number of times each toy, i, appears. The total uncertainty as a result of the statistical uncertainty on each sample is given by

$$(\sigma_p)^2 = \sum_i \left(\frac{\partial p}{\partial n_i} \sigma_{n_i} \right) \tag{12}$$

Expanding and using the Poisson uncertainty on the toys $\sigma_{n_i} = \sqrt{n_i}$ before taking the limit where each toy appears only once gives

$$(\sigma_p)^2 = \frac{p(1-p)}{N_{tot}} \tag{13}$$

Where N_{tot} is the integral of the test-statistic distribution. This is the expected result for a binomial process with a success rate p.

2.6. Bayes Factor

As in this analysis the $\Delta\chi^2$ statistic considered is marginalised over all nuisance parameters, the statistic can be interpreted in a Bayesian framework as a Bayes factor. The Bayes factor is another statistic (fully defined by the rate+shape analysis $\Delta\chi^2$ statistic) and gives an indication of which hypothesis gives the best fit to the data. The Bayes factor is defined as the marginal likelihood ratio of the two hypotheses in question.

$$B_{01} = \frac{e^{-\chi^2_{\beta=0}/2}}{e^{-\chi^2_{\beta=1}/2}} = e^{-\frac{1}{2}\Delta\chi^2} \tag{14}$$

The Jeffreys scale [9] is used to interpret the Bayes factor to show whether there is a preference for one hypothesis over the other. A more detailed discussion of Bayes factors can be found in [10].

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	175.69877	0.00417	11.13658	0.00020	0.02258	0.00015	186.86245
CC1pi	27.64351	0.00201	2.64220	0.00009	0.02418	0.00007	30.31206
CCcoherent	0.28884	0.00000	0.09306	0.00000	0.00000	0.00000	0.38190
2p2h	35.49529	0.00154	1.42670	0.00008	0.02578	0.00004	36.94944
CCother	5.49849	0.00087	0.41118	0.00008	0.00011	0.00001	5.91073
NC1pi0	0.61317	0.01856	0.02142	0.00145	N/A	N/A	0.65460
NC1piPM	4.89912	0.09882	0.17708	0.00990	N/A	N/A	5.18492
NCcoherent	0.00000	0.00000	0.00044	0.00004	N/A	N/A	0.00047
NCother	2.06264	0.07441	0.13205	0.00926	N/A	N/A	2.27837
Total	252.19982	0.20038	16.04070	0.02110	0.07266	0.00027	268.53493

Table 3: Predicted number of events in the neutrino mode single-ring μ-like sample obtained after applying the BANFF weight for a total exposure of 1.4938×10^{21} FHC POT. The Asimov data set A in Tab 2 is used. The total of 268.53493 events consists of 252.19982 signal events and 16.33511 background events.

3. Sensitivity to $\bar{\nu}_e$ Appearance

Since there are no fitted parameters in the analysis, the validation can be performed by checking for agreement between the oscillation analyses from both the VALOR and P-theta groups. This is done by performing a sensitivity study to $\bar{\nu}_e$ appearance data given the Asimov A data set, which is treated as we would treat the data. The Asimov A data set consists of the most probable data set (obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [6]) that is the nominal expected E_{reco} vs θ distribution [11] and the corresponding parameters are listed in Tab 2. The expected event rates for all five Super-K event samples using Asimov A parameters are shown in Tables 3 to 7.

Parameter(s)	Asimov A
$\sin^2\theta_{23}$	0.528
$\sin^2\theta_{13}$ reactors	0.0219
$\sin^2\theta_{12}$	0.304
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	2.509×10^{-3} eV^2/c^4
Δm^2_{21}	7.53×10^{-5} eV^2/c^4
δ_{CP}	-1.601
Mass Hierarchy	Normal
β	1

Table 2: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies corresponding to the Asimov A data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [4], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [6]. The mass hierarchy is not marginalised but fixed to either NH or IH.

The significance to $\bar{\nu}_e$ appearance is quoted by producing p-values using the procedure described in Sec 2. The suitability of the 20k toys used in this analysis is assessed by considering stability of the p-values, in appendix B. The expected Run 1-9c statistics sensitivity studies were performed for 1.4938×10^{21} POT and 1.1236×10^{21} POT in neutrino mode and antineutrino mode respectively.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.15943	6.36106	0.00638	0.25944	46.48200	0.28047	53.54879
CC1pi	0.04653	0.93873	0.00282	0.06851	4.48395	0.04775	5.58828
CCcoherent	0.00011	0.00836	0.00018	0.00403	0.03441	0.00330	0.05039
2p2h	0.02985	1.60917	0.00118	0.04339	8.69070	0.03920	10.41351
CCother	0.01337	0.12912	0.00035	0.00957	0.08761	0.00328	0.24329
NC1pi0	1.69779	0.03737	0.06073	0.00362	N/A	N/A	1.79950
NC1piPM	0.16808	0.00476	0.01032	0.00058	N/A	N/A	0.18374
NCcoherent	0.52906	0.00721	0.04837	0.00298	N/A	N/A	0.58763
NCother	0.31903	0.01710	0.02145	0.00130	N/A	N/A	0.35888
NC1gamma	0.94061	0.01701	0.05029	0.00237	N/A	N/A	1.01028
Total	3.90385	9.12991	0.20207	0.39579	59.77868	0.37400	73.78429

Table 4: Predicted number of events in the neutrino mode single-ring e-like sample obtained after applying the BANFF weight for a total exposure of 1.4938×10^{21} FHC POT. The Asimov data set A in Tab 2 is used. The total of 73.78429 events consists of 59.77868 signal events and 14.00562 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	22.71950	0.00126	44.41840	0.00138	0.00140	0.00204	67.14397
CC1pi	5.87728	0.00050	6.74025	0.00037	0.00089	0.00091	12.62020
CCcoherent	0.06995	0.00000	0.26113	0.00007	0.00000	0.00009	0.33125
2p2h	6.64836	0.00038	4.23289	0.00040	0.00065	0.00031	10.88300
CCother	1.40166	0.00018	0.90187	0.00000	0.00000	0.00000	2.30370
NC1pi0	0.08919	0.00330	0.09127	0.00251	N/A	N/A	0.18627
NC1piPM	0.53424	0.02270	0.68366	0.01835	N/A	N/A	1.25895
NCcoherent	0.00000	0.00000	0.00299	0.00000	N/A	N/A	0.00299
NCother	0.46952	0.02440	0.29873	0.01173	N/A	N/A	0.80438
Total	37.80970	0.05271	57.63119	0.03482	0.00294	0.00335	95.53471

Table 5: Predicted number of events in the antineutrino mode single-ring μ-like sample obtained after applying the BANFF weight for a total exposure of 1.1236×10^{21} RHC POT. The Asimov data set A in Tab 2 is used. The total of 95.53471 events consists of 57.63119 signal events and 37.90353 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.01912	0.75719	0.03547	1.25946	1.52367	4.23540	7.83030
CC1pi	0.01132	0.15029	0.01089	0.21446	0.21281	0.49728	1.09705
CCcoherent	0.00004	0.00148	0.00062	0.01585	0.00169	0.04650	0.06618
2p2h	0.00437	0.21366	0.00326	0.16792	0.34562	0.49049	1.22533
CCother	0.00578	0.03146	0.00178	0.01716	0.01547	0.00993	0.08158
NC1pi0	0.20578	0.00804	0.26242	0.00695	N/A	N/A	0.48317
NC1piPM	0.03677	0.00189	0.02971	0.00120	N/A	N/A	0.06958
NCcoherent	0.07106	0.00329	0.30096	0.00472	N/A	N/A	0.38002
NCother	0.08422	0.00327	0.03969	0.00147	N/A	N/A	0.12865
NC1gamma	0.11798	0.00653	0.25971	0.00570	N/A	N/A	0.38992
Total	0.55643	1.17710	0.94452	1.69488	2.09927	5.27959	11.75180

Table 6: Predicted number of events in the antineutrino mode single-ring e-like sample obtained after applying the BANFF weight for a total exposure of 1.1236×10^{21} RHC POT. The Asimov data set A in Tab 2 is used. The total of 11.75180 events consists of 5.27959 signal events and 6.47220 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.02845	0.02467	0.00118	0.00177	0.21197	0.00268	0.27071
CC1pi	0.07487	0.76673	0.00258	0.00389	4.85230	0.00229	5.70266
CCcoherent	0.00026	0.01781	0.00008	0.00007	0.11156	0.00011	0.12989
2p2h	0.01000	0.02214	0.00049	0.00096	0.15884	0.00106	0.19349
CCother	0.03763	0.11311	0.00117	0.00284	0.11952	0.00151	0.27577
NC1pi0	0.02039	0.00076	0.00084	0.00008	N/A	N/A	0.02208
NC1piPM	0.06503	0.00166	0.00386	0.00033	N/A	N/A	0.07089
NCother	0.20699	0.00838	0.01319	0.00090	N/A	N/A	0.22945
NC1gamma	0.02853	0.00000	0.00068	0.00007	N/A	N/A	0.02928
Total	0.47215	0.95526	0.02408	0.01091	5.45418	0.00764	6.92422

Table 7: Predicted number of events in the neutrino mode ν_eCC1π^+-like sample obtained after applying the BANFF weight for a total exposure of 1.4938×10^{21} FHC POT. The Asimov data set A in Tab 2 is used. The total of 6.92422 events consists of 5.45418 signal events and 1.47004 background events.

β	Analysis	p-value	σ	Asimov A Expectation
0	rate-only	0.036 ± 0.001	2.07 ± 0.01	N.A.
	rate+shape	0.0128 ± 0.0008	2.49 ± 0.02	
1	rate-only	0.386 ± 0.003	0.87 ± 0.01	$N_{events} = 11.8 \pm 0.9$
	rate+shape	0.426 ± 0.003	0.80 ± 0.01	$\Delta\chi^2 = 4.592$

Table 8: Asimov A p-values and event-rates for the $\beta = 0$ and $\beta = 1$ hypotheses using both the rate-only and rate+shape analyses. There are no Asimov A expectations for the $\beta = 0$ hypothesis, as the Asimov A dataset assumes that $\beta = 1$ is true.

The distributions of the event-rate in the antineutrino mode single-ring e-like sample for both hypotheses are shown in Fig 1 and the $\Delta\chi^2$ distributions for both hypotheses are shown in Fig 2. The corresponding Asimov A p-values are shown in Tab 8.

From the study of the Asimov data set A, which expects approximately 5 signal and 6 background events in the antineutrino mode e-like sample, we expect a p-value of about $1 - 2\,\%$ for the $\beta = 0$ hypothesis, corresponding to an exclusion of about $2.3 - 2.6\,\sigma$, where the conversion to σ is done using equation 15. We find a very large p-value of about $30\,\%$ in both analyses for the $\beta = 1$ hypothesis, corresponding to an exclusion of $< 1\,\sigma$, which is expected as Asimov data set A was generated using the $\beta = 1$ hypothesis. This is supported by the Bayes factor, which was found to be 0.101, indicating a substantial preference for the $\beta = 1$ hypothesis over the $\beta = 0$ hypothesis. The P-theta rate-only p-values for the Asimov A sensitivity study are 0.0383 and 0.373 for $\beta = 0$ and $\beta = 1$, respectively and the rate+shape p-values (using $E_{reco} - \theta$) are 0.0147 and 0.407 for $\beta = 0$ and $\beta = 1$, respectively [8]. The p-values produced by the VALOR and P-theta groups are considered to be within acceptable agreement. During the Run1-9c analysis, the reason for the remaining differences between the VALOR and P-theta p-values were thought to be due to differences in the toy generation methods. A comparison of the toy generation methods is shown in appendix C using full Run 1-9 statistics.

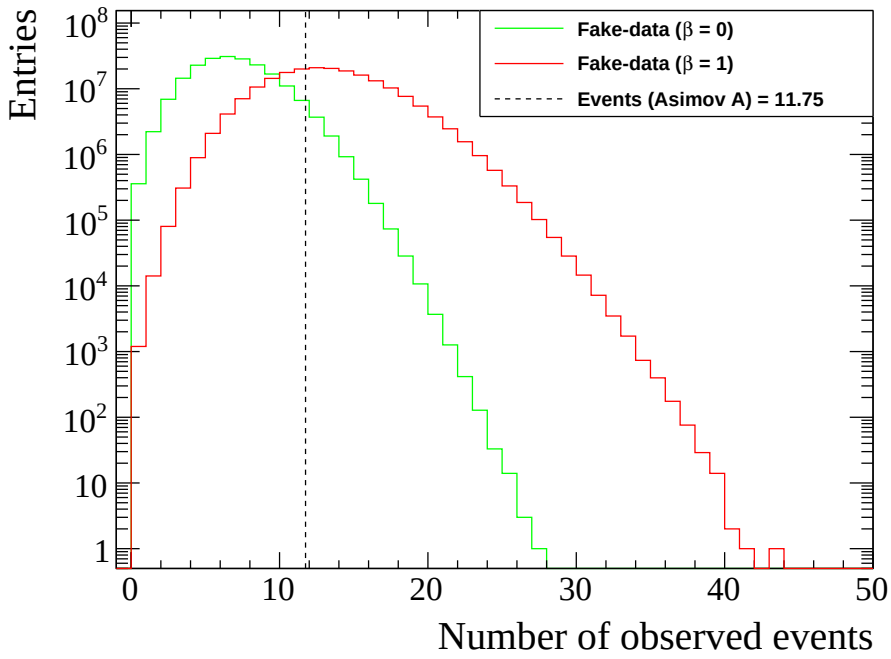


Figure 1: The distribution of the total number of events in the antineutrino mode single-ring e-like sample from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.1 using the Asimov A spectra to calculate the 4-sample likelihood. The dashed line corresponds to the total number of events given by the Asimov A data set.

$$\sigma = \sqrt{2}\,\mathrm{erf}^{-1}\,(1-p) \tag{15}$$

4. Results of the $\bar{\nu}_e$ Appearance Analysis

The rate-only and rate+shape p-values for $\bar{\nu}_e$ appearance are performed on the combined Run 1-9c statistics following the same procedure as in Sec 2. 1-sided p-values are computed on the right tail of the test statistic expected distribution to exclude $\beta = 0$ and the left tail to exclude $\beta = 1$. Fig 3 shows the data in the 5 Super-K event samples overlaid with the expected event rates for Asimov data set A. A summary of the observed number of events in Super-K, $\Delta\chi^2$ and p-values is given in Tab 9.

In Figs 4 and 5 the distributions of the test statistic, for rate-only and rate+shape analyses, are shown and compared with the data. Tab 9 contains a summary of the results. The rate-only analysis disfavours each hypothesis almost equally, whilst the rate+shape analysis disfavours the $\beta = 1$ hypothesis more than the $\beta = 0$ hypothesis. The Jeffreys scale indicates that the Bayes factor of 2.12 corresponds to a poor preference for the $\beta = 0$ hypothesis over $\beta = 1$. In comparison to the Run 1-7 analysis, the rate-only $\beta = 0$ p-value has decreased by a factor of 0.51. This is due to the number of observed events per POT increasing by a factor of 1.5, increasing the amount that the $\beta = 0$ hypothesis is disfavoured. The rate+shape $\beta = 0$ p-value has decreased by a factor of 0.57.

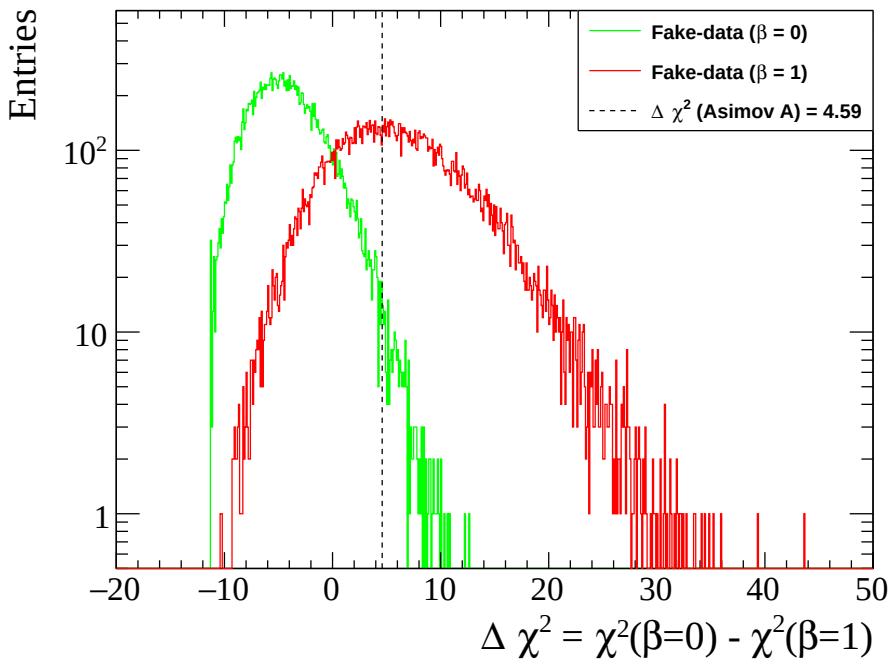
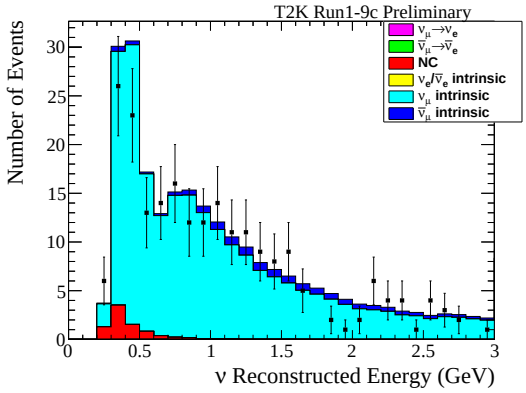


Figure 2: The distribution of $\Delta\chi^2$ values from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.2 using the Asimov A spectra to calculate the 4-sample likelihood. The dashed line corresponds to the $\Delta\chi^2$ value given by the Asimov A data set.

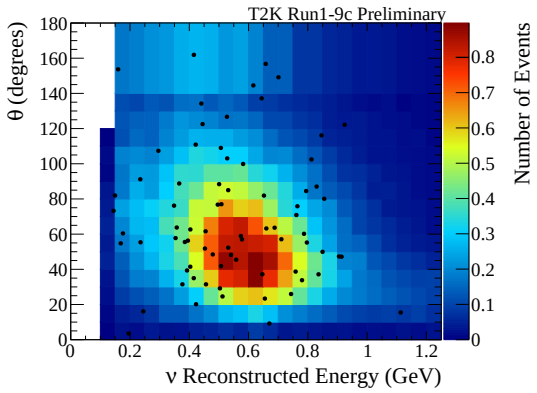
In addition to the number of observed events per POT increasing by a factor of 1.5, this can be explained by an increase in the number of events (compared to the Run 1-7 analysis [1]) in regions of the spectra in Fig 3d where a larger number of events are expected.

The P-theta rate-only p-values for the Run 1-9c data set are 0.219 and 0.213 for $\beta = 0$ and $\beta = 1$, respectively and the rate+shape p-values (using $E_{reco} - \theta$) are 0.214 and 0.0843 for $\beta = 0$ and $\beta = 1$, respectively [8]. The p-values produced by the VALOR and P-theta groups are considered to be within acceptable agreement. During the Run1-9c analysis, the reason for the remaining differences between the VALOR and P-theta p-values were thought to be due to differences in the toy generation methods. A comparison of the toy generation methods is shown in appendix C using full Run 1-9 statistics.

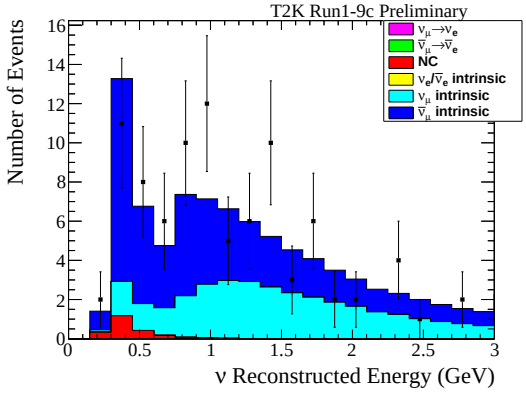
By breaking the electron-like antineutrino mode spectra from Fig 3 into its components, as shown in Fig 6, some explanation can be offered for the preference for the no $\bar{\nu}_e$ appearance hypothesis. The observed events do not appear to fit the Asimov A expected $\bar{\nu}_\mu \to \bar{\nu}_e$ distribution well, instead they are more consistent with the $\nu_\mu \to \nu_e$ and intrinsic ν_e & $\bar{\nu}_e$ backgrounds. As the shape of the spectra discredits the $\beta = 1$ hypothesis (which is information that the rate-only analysis does not have access to), the $\beta = 1$ p-value for the rate+shape analysis is substantially smaller than that of the rate-only analysis.



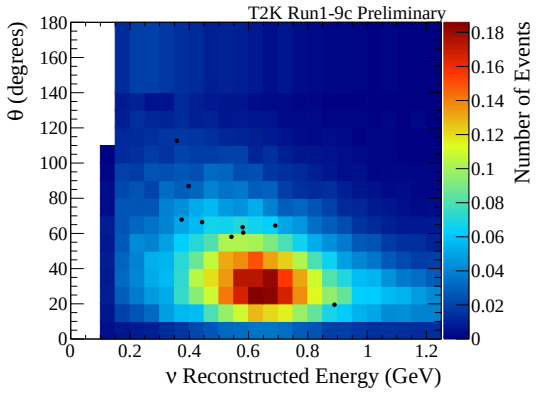
(a) Muon-like, neutrino mode



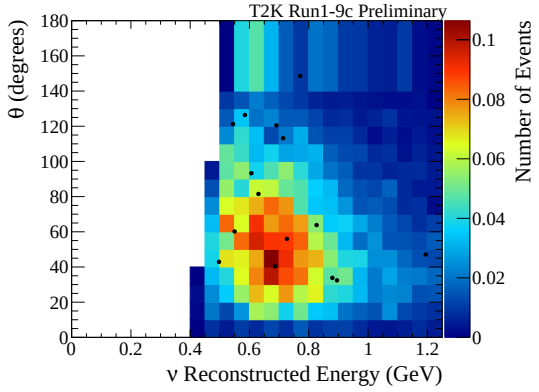
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like antineutrino mode



(e) ν_e CC1π^+-like, neutrino mode

Figure 3: Predicted Asimov A spectra. μ-like distributions are a function of the reconstructed neutrino energy, while the e-like, including ν_e CC1π^+, distributions are functions of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full Run 1-9c data set. The spectra are generated with the oscillation parameters corresponding to the Asimov data set A shown in table 2.

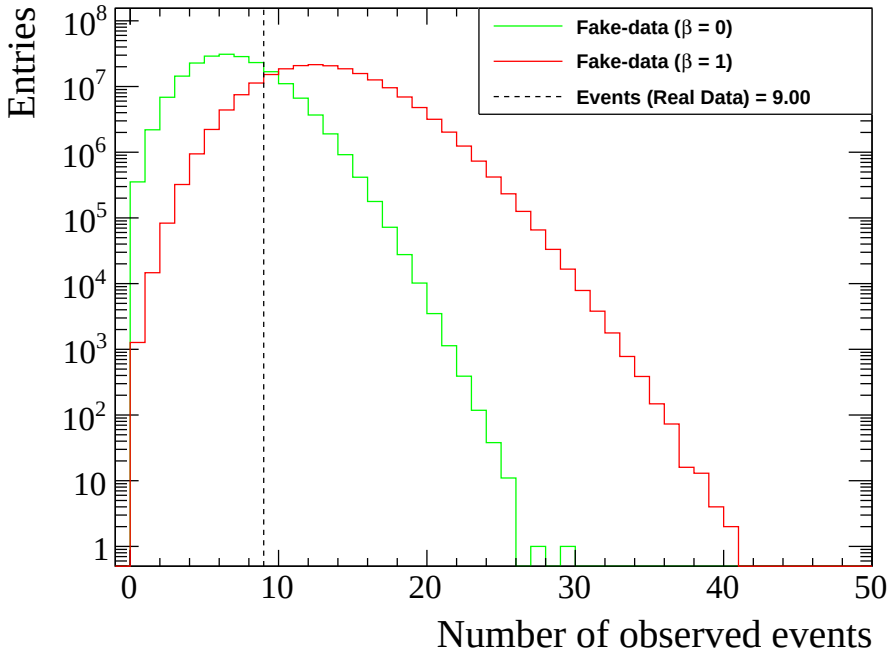


Figure 4: The distribution of the total number of events in the antineutrino mode single-ring e-like sample from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.1 using the Run 1-9c data spectra to calculate the 4-sample likelihood. The dashed line corresponds to the total number of events given by the Run 1-9c data set.

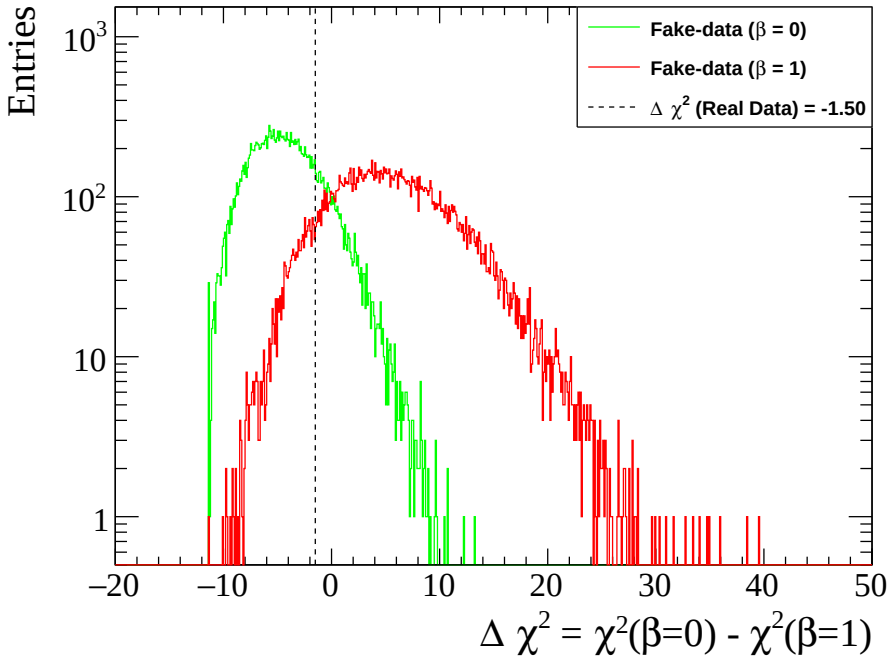
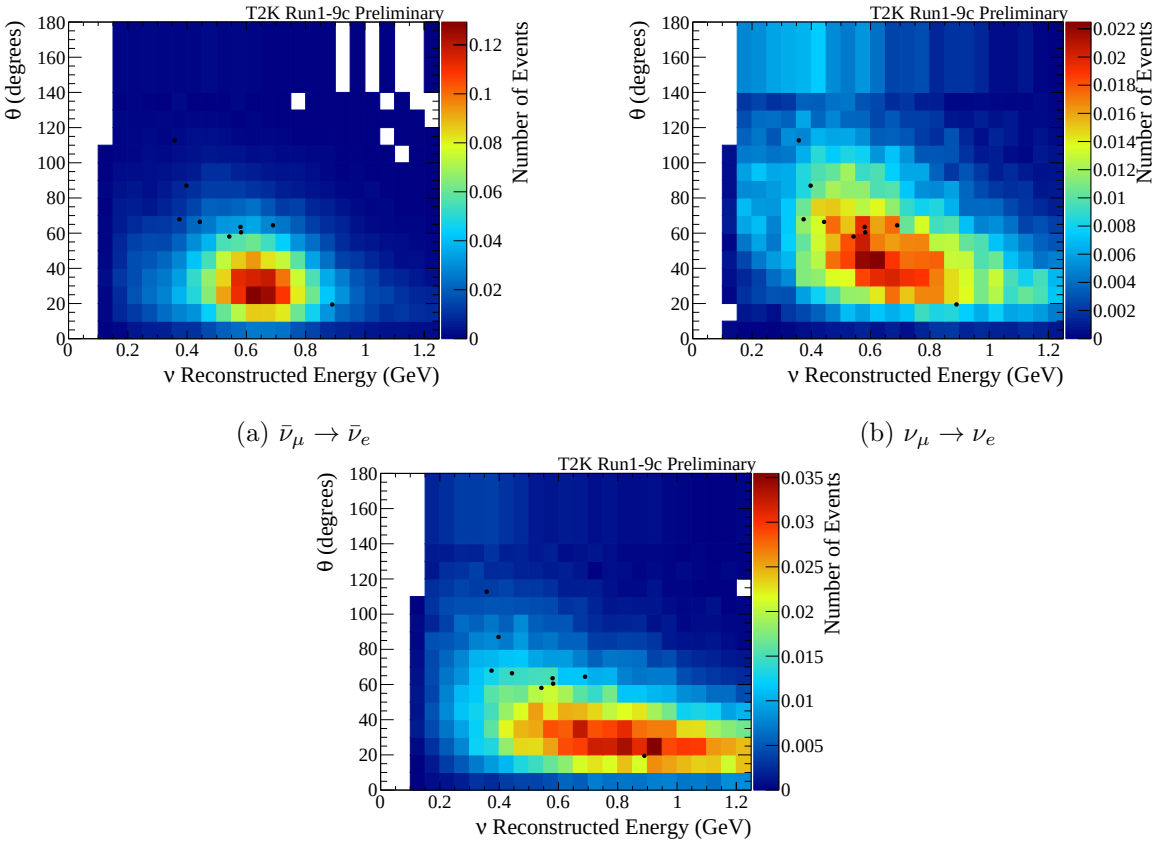


Figure 5: The distribution of $\Delta\chi^2$ values from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.2 using the Run 1-9c data spectra to calculate the 4-sample likelihood. The dashed line corresponds to the $\Delta\chi^2$ value given by the Run 1-9c data set.



(a) $\bar{\nu}_\mu \to \bar{\nu}_e$

(b) $\nu_\mu \to \nu_e$

(c) Intrinsic ν_e + Intrinsic $\bar{\nu}_e$

Figure 6: Predicted Asimov A Electron-like antineutrino mode spectra broken down by component. The distributions correspond to the statistics collected in the full Run 1-9c data set. The spectra are generated with the oscillation parameters corresponding to the Asimov data set A shown in table 2.

Run	RHC POT	Rate-only Analysis			Rate+shape Analysis			
		N_{obs}	$\beta = 0$ p-value	$\beta = 1$ p-value	$\Delta\chi^2_{\text{obs}}$	$\beta = 0$ p-value	$\beta = 1$ p-value	B_{01}
1-9c	1.1236×10^{21}	9	0.209 ± 0.003	0.210 ± 0.003	-1.502	0.215 ± 0.003	0.092 ± 0.003	2.12
1-7	7.471×10^{20}	4	0.410	0.2155	-1.930	0.3742	0.099	2.62

Table 9: Summary of observed number of events, $\Delta\chi^2$, rate-only and rate+shape p-values and Bayes factor from the combined Run 1-9c data set compared to the Run 1-7 data set used for the previous iteration of this analysis, in 2016 [1].

5. Summary

In this technical note, we have presented the results of the $\bar{\nu}_e$ appearance analysis using the 3-flavour $\nu/\bar{\nu}$ joint analysis framework performed by the VALOR group, on the combined Run 1-9c data set, for an exposure of 1.4938×10^{21} POT in neutrino mode and 1.1236×10^{21} POT in antineutrino mode. The goal of the analysis is to search for $\bar{\nu}_e$ appearance in a $\bar{\nu}_\mu$ beam.

This analysis predicts 268.5 ± 13.2 (syst) for neutrino μ-like, 95.5 ± 4.1 (syst) for antineutrino μ-like, 73.8 ± 7.1 (syst) for neutrino e-like, 11.8 ± 0.9 (syst) for antineutrino e-like and 6.9 ± 1.3 (syst) for neutrino ν_eCC1π^+-like events in Super K and observes respectively 243 for neutrino μ-like, 102 for antineutrino μ-like, 75 for neutrino e-like, 9 for antineutrino e-like and 15 for neutrino ν_eCC1π^+-like events.

Both a rate only and a rate+shape analysis are performed. In both analyses, fits were performed in the joint fit framework, including all samples as data to constraint either background and the hypotheses tested. The number of observed events in the antineutrino mode e-like spectrum is 9 and the measured rate+shape test statistic, $\Delta\chi^2 = \chi^2\left(\beta = 0\right) - \chi^2\left(\beta = 1\right)$ was found to be -1.502. The significance of the result is shown as a p-value. Taking no $\bar{\nu}_e$ appearance as our null hypothesis we find a p-value of 0.209 ± 0.003 for the rate-only analysis and a p-value of 0.215 ± 0.003 for the rate+shape analysis. Taking $\bar{\nu}_e$ appearance according to the best knowledge of the PMNS matrix as our null hypothesis we find a p-value of 0.210 ± 0.003 for the rate-only analysis and a p-value of 0.092 ± 0.003 for the rate+shape analysis. The Bayes factor, given by the rate+shape analysis was found to be 2.12 indicating a poor preference for no $\bar{\nu}_e$ appearance.

Parameter(s)	Prior	Range
$\sin^2\theta_{23}$	Uniform	$[0.3; 0.7]$
$\sin^2 2\theta_{13}$ reactors	Gaussian	0.0830 ± 0.0031
$\sin^2 2\theta_{12}$	Gaussian	0.846 ± 0.021
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{13}\rvert$ (IH)	Uniform	$[2.3; 2.7] \times 10^{-3}$ eV^2/c^4
Δm^2_{21}	Gaussian	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4
δ_{CP}	Uniform	$[-\pi; +\pi]$
Mass Hierarchy	Bernoulli	0.5 normal, 0.5 inverted

Table 10: Treatment of the oscillation parameters in the $\bar{\nu}_e$ appearance analysis. All the gaussian priors are from [5]. The parameter $\sin^2 2\theta_{13}$ is constrained with the measurement of reactor experiments using the gaussian prior. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [5], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [6].

6. Updated Analysis Using the Full T2K Run 1-9 Dataset

A statistical update to the Run 1-9c results has been performed with the addition of the Run 9d dataset. The analysis procedure is identical to that of the Run 1-9c analysis, but there has been the following changes to the analysis inputs:

- An increase of $\bar{\nu}$ mode exposure of 46 % from 1.1236×10^{21} POT to 1.6346×10^{21} POT
- Updated the Gaussian reactor constraint on $\sin^2 2\theta_{13}$ from the 2016 PDG value of 0.0857 ± 0.0046 to the 2018 PDG value of 0.0830 ± 0.0031, as shown in the updated prior table 10
- Updated the Asimov A true value of $\sin^2 2\theta_{13}$ to 0.0830, to match the central value of the above 2018 PDG reactor constraint
- An updated flux tuning using an updated version of FLUKA and nuclear interaction model [14]

As part of the validation process two additional studies have been performed: An updated test of the p-value stability (see appendix B.1), and a comparison of the toy generation methods used by the VALOR and P-theta groups (see appendix C).

6.1. Sensitivity to $\bar{\nu}_e$ Appearance

The sensitivity study is performed as described for the Run 1-9c analysis (see sec 3). The expected event rates for all five Super-K event samples using the updated Asimov A parameters are shown in Tables 11 to 15. Although there has been no change in the neutrino mode exposure, the event rates in the neutrino mode samples have changed slightly compared to the Run 1-9c values due to the updated flux tuning.

The distributions of the event-rate in the antineutrino mode single-ring e-like sample for both hypotheses are shown in Fig 7 and the $\Delta\chi^2$ distributions for both hypotheses are shown in Fig 8. The corresponding Asimov A p-values are shown in Tab 16. A comparison to the sensitivity study produced by the P-theta group is shown in appendix E.1.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	178.45275	0.00420	11.18602	0.00019	0.02219	0.00014	189.66550
CC1pi	28.23407	0.00203	2.65612	0.00009	0.02380	0.00007	30.91619
CCcoherent	0.29228	0.00000	0.09292	0.00000	0.00000	0.00000	0.38520
2p2h	35.80423	0.00153	1.44684	0.00008	0.02483	0.00004	37.27755
CCother	5.52439	0.00086	0.40845	0.00008	0.00011	0.00001	5.93389
NC1pi0	0.62495	0.01869	0.02138	0.00143	N/A	N/A	0.66646
NC1piPM	4.99423	0.09960	0.17726	0.00980	N/A	N/A	5.28090
NCcoherent	0.00000	0.00000	0.00044	0.00003	N/A	N/A	0.00048
NCother	2.10279	0.07529	0.13268	0.00923	N/A	N/A	2.31998
Total	256.02970	0.20221	16.12211	0.02094	0.07094	0.00026	272.44616

Table 11: Predicted number of events in the neutrino mode single-ring μ-like sample obtained after applying the BANFF weight for a total exposure of 1.4938×10^{21} FHC POT. The Asimov data set A in Tab 2 is used. The total of 272.44616 events consists of 256.02970 signal events and 16.41647 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.16374	6.41338	0.00641	0.25515	45.62597	0.27069	52.73533
CC1pi	0.04802	0.95057	0.00284	0.06769	4.44444	0.04635	5.55993
CCcoherent	0.00012	0.00842	0.00018	0.00394	0.03387	0.00317	0.04970
2p2h	0.03051	1.60651	0.00120	0.04313	8.46160	0.03827	10.18123
CCother	0.01340	0.12849	0.00034	0.00930	0.08589	0.00313	0.24056
NC1pi0	1.72930	0.03761	0.06069	0.00357	N/A	N/A	1.83117
NC1piPM	0.17178	0.00480	0.01033	0.00057	N/A	N/A	0.18749
NCcoherent	0.53883	0.00729	0.04868	0.00296	N/A	N/A	0.59776
NCother	0.32454	0.01730	0.02153	0.00129	N/A	N/A	0.36467
NC1gamma	0.96236	0.01722	0.05061	0.00235	N/A	N/A	1.03254
Total	3.98261	9.19160	0.20281	0.38996	58.65176	0.36161	72.78036

Table 12: Predicted number of events in the neutrino mode single-ring e-like sample obtained after applying the BANFF weight for a total exposure of 1.4938×10^{21} FHC POT. The Asimov data set A in Tab 2 is used. The total of 72.78036 events consists of 58.65176 signal events and 14.12860 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	33.23334	0.00183	64.88124	0.00199	0.00198	0.00287	98.12326
CC1pi	8.61735	0.00072	9.82622	0.00054	0.00126	0.00129	18.44737
CCcoherent	0.10172	0.00000	0.37877	0.00010	0.00000	0.00013	0.48073
2p2h	9.62489	0.00056	6.23791	0.00058	0.00091	0.00044	15.86529
CCother	2.03601	0.00025	1.28783	0.00000	0.00000	0.00000	3.32409
NC1pi0	0.13032	0.00479	0.13243	0.00358	N/A	N/A	0.27112
NC1piPM	0.78284	0.03295	0.99469	0.02624	N/A	N/A	1.83671
NCcoherent	0.00000	0.00000	0.00439	0.00000	N/A	N/A	0.00439
NCother	0.69330	0.03560	0.43187	0.01681	N/A	N/A	1.17759
Total	55.21978	0.07670	84.17534	0.04984	0.00416	0.00472	139.53053

Table 13: Predicted number of events in the antineutrino mode single-ring μ-like sample obtained after applying the BANFF weight for a total exposure of 1.6346×10^{21} RHC POT. The Asimov data set A in Tab 2 is used. The total of 139.53053 events consists of 84.17534 signal events and 55.35520 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.02796	1.10898	0.05220	1.82304	2.15856	5.96183	11.13258
CC1pi	0.01660	0.21958	0.01604	0.30984	0.30251	0.70366	1.56823
CCcoherent	0.00005	0.00214	0.00091	0.02278	0.00240	0.06516	0.09344
2p2h	0.00633	0.30850	0.00484	0.24505	0.48467	0.69725	1.74663
CCother	0.00835	0.04508	0.00253	0.02421	0.02166	0.01383	0.11566
NC1pi0	0.30059	0.01164	0.38057	0.00988	N/A	N/A	0.70269
NC1piPM	0.05384	0.00275	0.04314	0.00171	N/A	N/A	0.10143
NCcoherent	0.10435	0.00481	0.44089	0.00678	N/A	N/A	0.55682
NCother	0.12374	0.00477	0.05756	0.00212	N/A	N/A	0.18819
NC1gamma	0.17384	0.00959	0.38085	0.00822	N/A	N/A	0.57250
Total	0.81566	1.71784	1.37954	2.45362	2.96980	7.44172	16.77817

Table 14: Predicted number of events in the antineutrino mode single-ring e-like sample obtained after applying the BANFF weight for a total exposure of 1.6346×10^{21} RHC POT. The Asimov data set A in Tab 2 is used. The total of 16.77817 events consists of 7.44172 signal events and 9.33646 background events.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.02898	0.02486	0.00118	0.00173	0.20835	0.00258	0.26767
CC1pi	0.07743	0.77592	0.00261	0.00385	4.79666	0.00222	5.65868
CCcoherent	0.00027	0.01791	0.00008	0.00007	0.10930	0.00010	0.12773
2p2h	0.01020	0.02210	0.00050	0.00096	0.15575	0.00103	0.19054
CCother	0.03776	0.11255	0.00115	0.00275	0.11687	0.00144	0.27253
NC1pi0	0.02080	0.00077	0.00084	0.00008	N/A	N/A	0.02249
NC1piPM	0.06592	0.00167	0.00387	0.00033	N/A	N/A	0.07179
NCother	0.21033	0.00847	0.01326	0.00090	N/A	N/A	0.23296
NC1gamma	0.02908	0.00000	0.00068	0.00007	N/A	N/A	0.02983
Total	0.48077	0.96425	0.02418	0.01074	5.38692	0.00737	6.87423

Table 15: Predicted number of events in the neutrino mode ν_eCC1π^+-like sample obtained after applying the BANFF weight for a total exposure of 1.4938×10^{21} FHC POT. The Asimov data set A in Tab 2 is used. The total of 6.87423 events consists of 5.38692 signal events and 1.48730 background events.

β	Analysis	p-value	σ	Asimov A Test Statistic
0	rate-only	0.0185 ± 0.0009	2.36 ± 0.02	N.A.
	rate+shape	0.0057 ± 0.0005	2.76 ± 0.03	
1	rate-only	0.379 ± 0.003	0.88 ± 0.01	$N_{events} = 16.8 \pm 1.3$ (syst)
	rate+shape	0.409 ± 0.004	0.83 ± 0.01	$\Delta\chi^2 = 6.27$

Table 16: Asimov A p-values and event-rates for the $\beta = 0$ and $\beta = 1$ hypotheses using both the rate-only and rate+shape analyses and the full Run 1 - 9 dataset. There are no Asimov A expectations for the $\beta = 0$ hypothesis, as the Asimov A dataset assumes that $\beta = 1$ is true.

For both rate-only and rate+shape analyses, the Asimov A sensitivity using the full Run 1-9 dataset gives a $\sim 0.3\,\sigma$ stronger exclusion of the $\beta = 0$ hypothesis compared to the sensitivity using the Run 1-9c dataset. There is approximately no change in the exclusion of the $\beta = 1$ hypothesis. A more detailed comparison of the full Run 1-9 results to the Run 1-9c results can be found in appendix D.

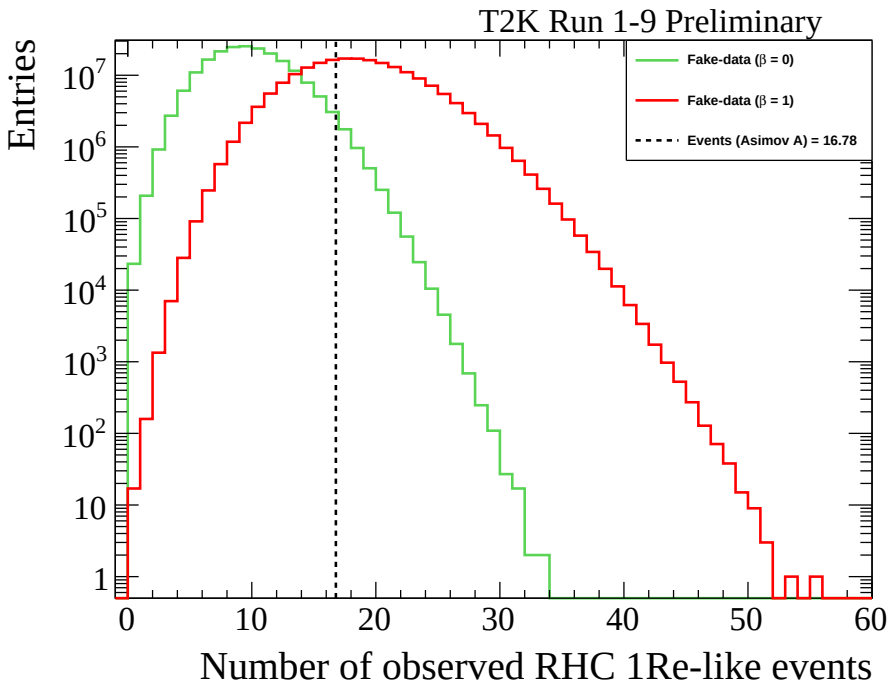


Figure 7: The distribution of the total number of events in the antineutrino mode single-ring e-like sample from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses using full Run 1 - 9 statistics. The distributions were produced using the method outlined in section 2.1 using the Asimov A spectra to calculate the 4-sample likelihood. The dashed line corresponds to the total number of events given by the Asimov A data set.

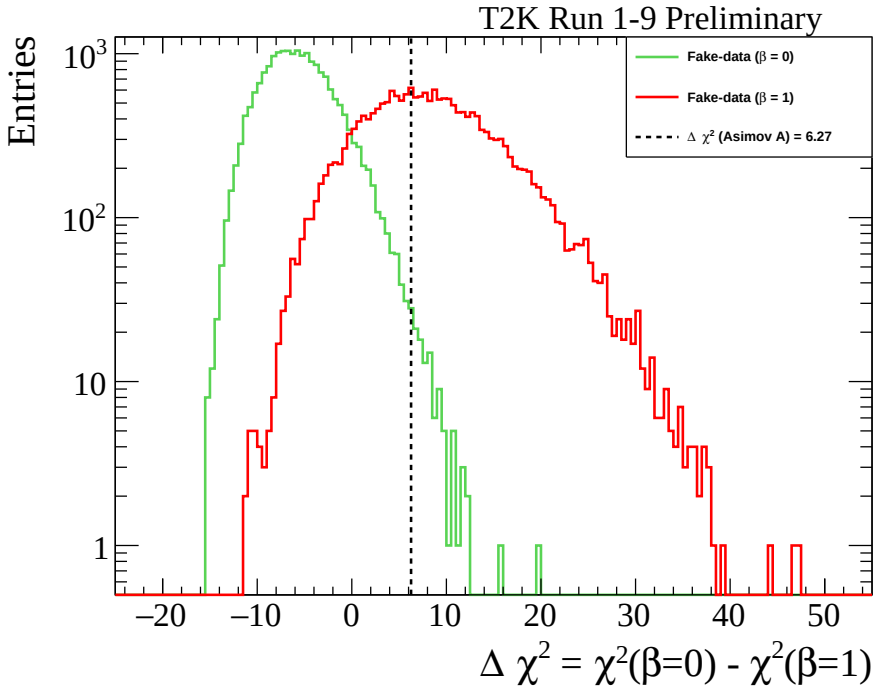


Figure 8: The distribution of $\Delta\chi^2$ values from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses using full Run 1 - 9 statistics. The distributions were produced using the method outlined in section 2.2 using the Asimov A spectra to calculate the 4-sample likelihood. The dashed line corresponds to the $\Delta\chi^2$ value given by the Asimov A data set.

6.2. Results of the $\bar{\nu}_e$ Appearance Analysis

The rate-only and rate+shape p-values for $\bar{\nu}_e$ appearance are performed on the full Run 1-9 data set following the same procedure as in Sec 2. 1-sided p-values are computed on the right tail of the test statistic expected distribution to exclude $\beta = 0$ and the left tail to exclude $\beta = 1$. Fig 11 shows the data in the 5 Super-K event samples overlaid with the expected event rates for Asimov data set A. In Figs 9 and 10 the distributions of the test statistic, for rate-only and rate+shape analyses, are shown and compared with the data. The observed number of events in the antineutrino mode single-ring e-like sample in Super-K was 15. The observed $\Delta\chi^2$ value was 3.811. The p-values resulting from these distributions and test statistic values are shown in Tab 17. A comparison to observed p-values produced by the P-theta group is shown in appendix E.2..

As shown in Tab 17, both the rate-only and rate+shape analyses disfavour the $\beta = 0$ hypothesis more than the $\beta = 1$ hypothesis. The Jeffreys scale indicates that the Bayes factor (see eq. 14) of 0.149 corresponds to a substantial preference for the $\beta = 1$ hypothesis over $\beta = 0$. This preference is in contrast to the earlier Run 1-9c and Run 1-7 analyses, which both had a poor preference for the $\beta = 0$ hypothesis over $\beta = 1$. The comparison between the full Run 1-9 and Run 1-9c analyses is discussed in more detail in Appendix D.

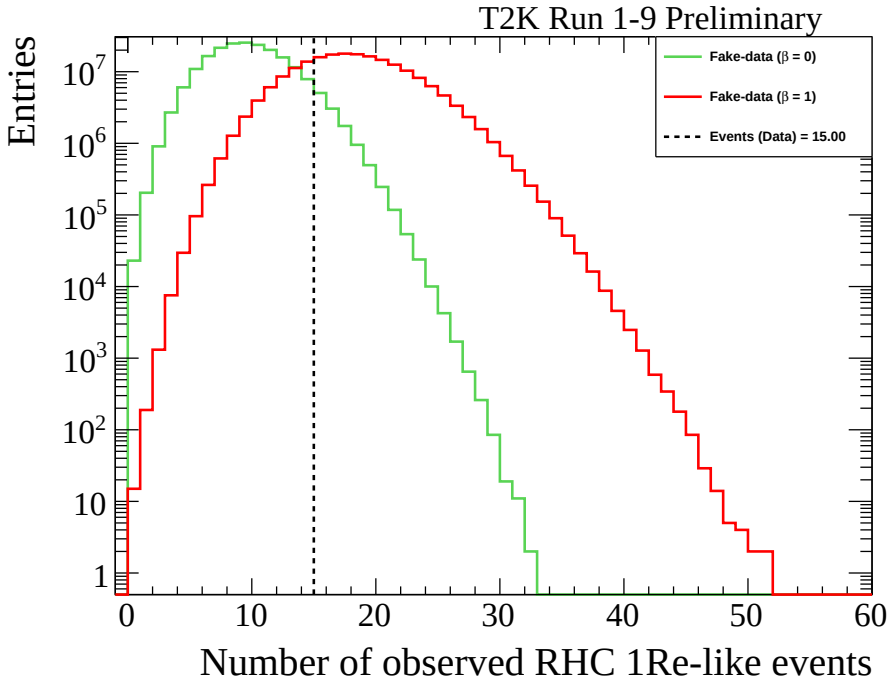


Figure 9: The distribution of the total number of events in the antineutrino mode single-ring e-like sample from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.1 using the Run 1-9d data spectra to calculate the likelihood. The dashed line corresponds to the total number of events given by the Run 1-9d data set.

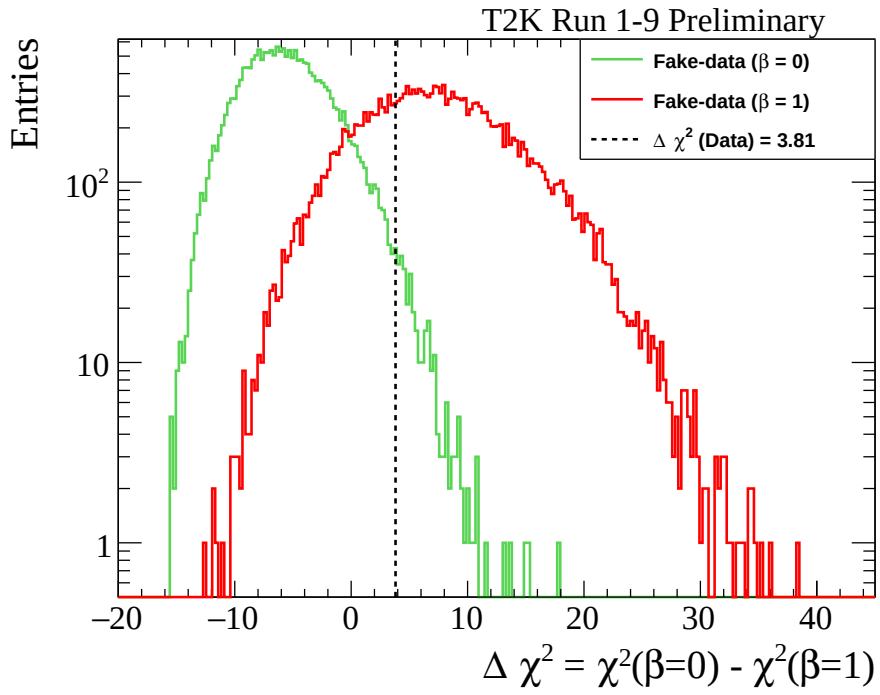
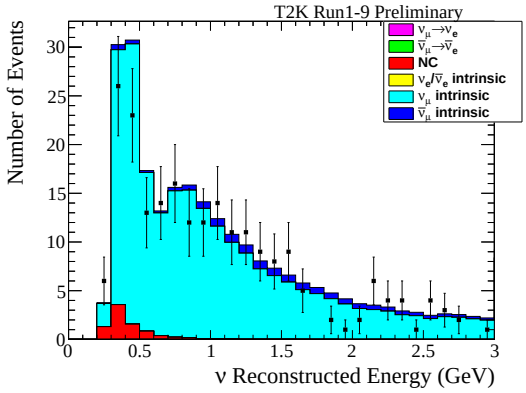


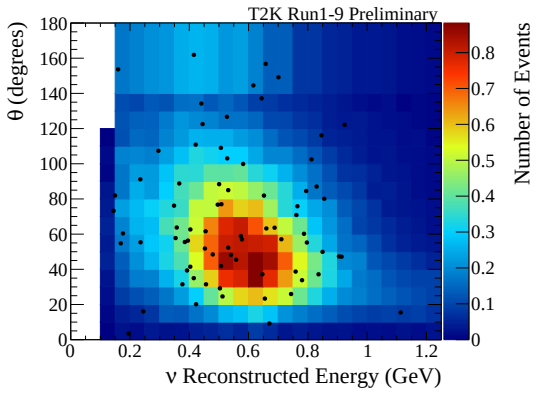
Figure 10: The distribution of $\Delta\chi^2$ values from each of the 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses. The distributions were produced using the method outlined in section 2.2 using the Run 1-9d data spectra to calculate the likelihood. The dashed line corresponds to the $\Delta\chi^2$ value given by the Run 1-9d data set.

β	Analysis	p-value	σ
0	rate-only	0.059 ± 0.002	$1.89^{+0.02}_{-0.01}$
	rate+shape	0.0163 ± 0.0009	2.40 ± 0.02
1	rate-only	0.321 ± 0.003	0.99 ± 0.01
	rate+shape	0.300 ± 0.004	1.04 ± 0.01

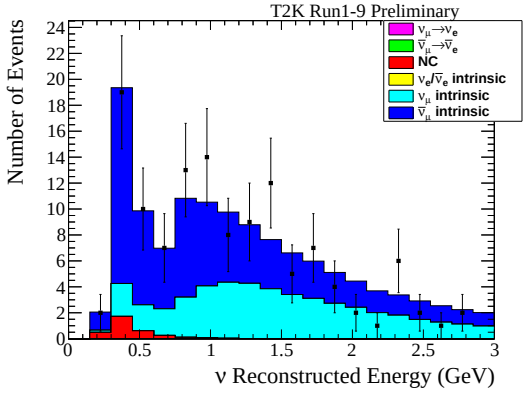
Table 17: Data p-values and event-rates for the $\beta = 0$ and $\beta = 1$ hypotheses using both the rate-only and rate+shape analyses and the full Run 1-9 dataset



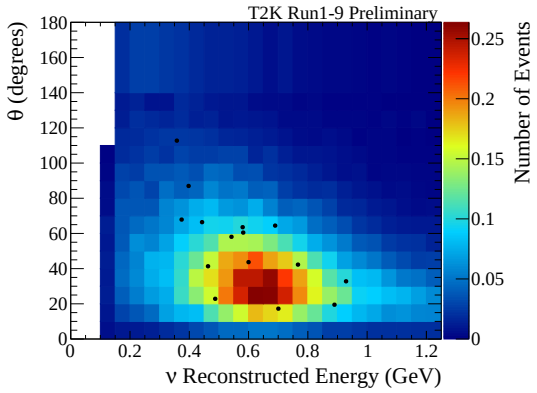
(a) Muon-like, neutrino mode



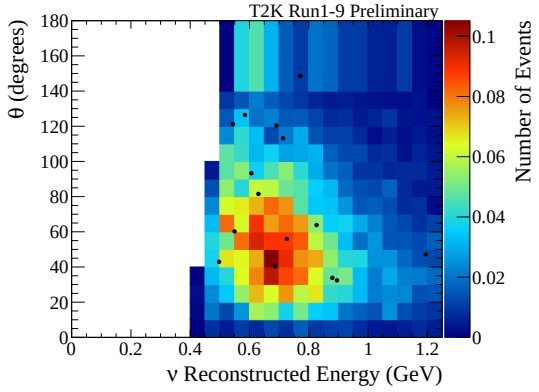
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like antineutrino mode



(e) ν_e CC1π^+-like, neutrino mode

Figure 11: Predicted Asimov A spectra with Run 1-9 data superimposed. μ-like distributions are a function of the reconstructed neutrino energy, while the e-like, including ν_e CC1π^+, distributions are functions of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full Run 1-9 data set. The spectra are generated with the oscillation parameters corresponding to the Asimov data set A shown in table 2.

[1] C.Andreopoulos et al., $\bar{\nu}_e$ Appearance Analysis with the 3-Flavour Joint Fit Framework for T2K 6.914×10^{20}-POT Neutrino 4.011×10^{20}-POT Anti-Neutrino T2K-TN-296, http://www.t2k.org/docs/technotes/296

[2] C.Andreopoulos et al., T2K Neutrino and Anti-Neutrino 3-Flavour Joint Analysis of Run 1-9c (1.4938×10^{21}-POT ν 1.1236×10^{21}-POT $\bar{\nu}$) data sets T2K-TN-327, http://www.t2k.org/docs/technotes/327

[3] C.Andreopoulos et al., T2K Neutrino and Anti-Neutrino 3-Flavour Joint Analysis of Run 1-9c (1.4938×10^{21}-POT ν 1.6346×10^{21}-POT $\bar{\nu}$) data sets - Statistical Update T2K-TN-360, https://t2k.org/docs/technotes/360/run19d/statistical-update

[4] K.A. Olive et al. (Particle Data Group 2014 with 2015 update), Chin. Phys. C, 38, 090001

[5] M. Tanabashi et al. (Particle Data Group), Phys. Rev. D 98, 030001 (2018) t2k neutrino

[6] T2K Collaboration (K.Abe et al.), Measurement of neutrino oscillation in appearance and disappearance channels by the T2K experiment with 6.6×10^{20} protons on target http://www.t2k.org/comm/pubboard/JournalPapers/oa2014prd/draft3

[7] S. Berkman, P. de Perio, A. Konaka, A. Missert, H. Tanaka, S. Tobayama, M. J. Wilking, E. D. Zimmerman fiTQun: A New Reconstruction Algorithm for Super-K http://www.t2k.org/docs/technotes/146

[8] T.Koga, S.Bienstock, C.Bronner and B.Quilain, Updated p-theta $\bar{\nu}_e$ appearance search using the T2K RUN1-9 data set, http://www.t2k.org/docs/technotes/353

[9] H. Jeffreys The Theory of Probability OUP Oxford 1998

[10] A. Kaboth, K. Duffy, R Calland, and D Payne Analysis of $\bar{\nu}_e$ Appearance using MCMC T2K-TN-247 http://www.t2k.org/docs/technotes/247

[11] G.Cowan, K.Cranmer, E.Gross, O.Vitells, "Asymptotic formulae for likelihood-based tests of new physics" arXiv:1007.1727v2

[12] S. Wilks, The Large-Sample Distribution of the Likelihood Ratio for Testing Composite Hypotheses, Ann. Math. Statist., 03, 1, 60-62, https://doi.org/10.1214/aoms/1177732360

[13] F. Grubbs, Sample Criteria for Testing Outlying Observations, Ann. Math. Statist., 03, 1, 27-58, https://doi.org/10.1214/aoms/1177729885

[14] T. Vladisavljevic, Full Run 9 T2K Flux Release of Run 9d (2018/3/9 2018/5/31), https://t2k.org/beam/NuFlux/FluxRelease/13arelease/13av3-tuned-fluxes-for-runs-1-9d-slides

[15] F. Bench, $\bar{\nu}_e$ Appearance Analysis, https://t2k.org/meet/collab/201805/talks/fripm/nuebar

Appendices

A. Optimisation of Oscillation Parameter Priors

When generating fake-data toys that are constrained by the Run 1-9c real-data (see section 2), it was found that the amount of CPU time required to generate these toys was $\sim 80\times$ the CPU time required to generate toys constrained by Asimov A fake-data.

The reason for this is that the constraints on the oscillation parameters given by the Run 1-9c data are substantially stronger than the Asimov A constraints [2]. This results in a larger fraction of the oscillation parameters being generated at more unlikely values and therefore the mean probability of accepting each fake-data toy decreased by a factor of 138.

The large increase in CPU time meant that the toy generation process would require an unreasonably large amount of time and the majority of fake-data toys generated would be immediately rejected. In order to increase the efficiency of the fake-data toy generation process, the oscillation parameter priors ranges were reduced from the ranges used to generate marginalization and fake-data toys in the Run 1-9c joint analysis [2], these ranges will be referred to as the *nominal priors* and are shown in the *Range for marginalisation toys* column of Tab 1.

By restricting the range of the each oscillation parameter prior to the region close to the Run 1-9c best-fit point [2], a larger fraction of fake-data toys generated will have a higher likelihood compared to toys generated using the nominal prior ranges (referred to as the *nominal set*), resulting in a reduction in the required CPU time. However, there is a danger that by reducing the prior range, a toy that would have been accepted from the nominal set would no longer be generated and so would be excluded from the new set of accepted toys. This could cause a change in the measured p-values, which would be undesirable.

To quantify this effect on the p-values, the maximum possible change in p-value is calculated as follows. Equations 5 and 6 for the calculation of the p-value can be re-written as

$$\text{p-value} = p = \frac{n}{N} \tag{16}$$

Where n is the number of accepted toys in the tail of interest and N is the total number of accepted toys in the test-statistic distribution. At worst, all toys that were excluded from the nominal set could have been in the tail of interest, but when using the reduced prior, these excluded toys would have been re-thrown (to keep the total number of accepted toys constant). At worst, all of these re-thrown toys would be outside of the tail of interest (or vice-versa). Using equation 16, this gives a maximum possible p-value change, $|\Delta p|_{\text{max}}$ of

$$|\Delta p|_{\text{max}} = |p_{\text{nominal}} - p| = \left| \frac{n_{\text{nominal}}}{N} - \frac{n_{\text{nominal}} \pm n_{\text{excluded}}}{N} \right| = \frac{n_{\text{excluded}}}{N} \tag{17}$$

Therefore, by reducing the prior ranges successively for a number of oscillation parameters and calculating $|\Delta p|_{\text{max}}$ for various potential prior ranges, the CPU time required can be minimised whilst keeping $|\Delta p|_{\text{max}}$ smaller than the minimum error on the p-values observed in the Asimov

A sensitivity study (see section 3), which was 8.0×10^{-4}. This is demonstrated for a number of different $\sin^2\theta_{23}$ prior ranges in Tab 18, from which a new prior of $[0.42, 0.62]$ is chosen. Using the same method with this new prior range, Δm^2_{23} and subsequently δ_{CP} have their prior ranges optimised, the results of which are shown in Tab 19.

$\sin^2\theta_{23}$ Prior range	Toys excluded from the nominal set (of 20,000 accepted toys)	Maximum possible p-value change	CPU time (current / nominal)
$[0.300, 0.700]$ (nominal)	0	0	1
$[0.355, 0.680]$	1.32×10^{-8}	6.64×10^{-13}	0.85
$[0.375, 0.660]$	2.40×10^{-5}	1.20×10^{-9}	0.71
$[0.390, 0.650]$	7.34×10^{-4}	3.97×10^{-8}	0.66
$[0.410, 0.625]$	1.25	6.24×10^{-5}	0.55
$[0.420, 0.620]$	5.04	2.52×10^{-4}	0.52
$[0.430, 0.610]$	52.5	2.62×10^{-3}	no data
$[0.440, 0.605]$	134	6.74×10^{-3}	no data

Table 18: Maximum possible effect on the p-values obtained by varying the $\sin^2 2\theta_{23}$ prior range used to generate toys to fill the test-statistic distributions. *Toys excluded from the nominal set* are the number of toys that would have been accepted (using the MC sampling method) if the nominal prior range had been used but lie outside of the range of the tighter prior under consideration. These are calculated assuming 20,000 accepted toys will be generated.

Parameter	Nominal Prior	Optimised Prior	Maximum possible p-value change (10^{-4})	CPU time (Optimised / Nominal)
$\sin^2\theta_{23}$	$[0.30, 0.70]$	$[0.42, 0.62]$	2.5	0.52
Δm^2_{23}	$[2.30, 2.70]$	$[2.30, 2.65]$	4.6	0.45
δ_{CP}	$[-\pi, \pi]$	$[-\pi, 1.1]$ & $[1.8, \pi]$	7.8	0.39

Table 19: Optimised priors for the oscillation parameters that are randomised according to flat prior probability distributions. The maximum possible effect on the p-values and the expected reduction on CPU time are shown. These are calculated assuming 20,000 accepted toys will be generated.

A.1. Updated Priors for the Full Run 1-9 Analysis

As shown by the full Run 1-9 sensitivity study (see Tab 16), the smallest p-value error has reduced from 8.0×10^{-4} to 5.3×10^{-4} so the prior ranges must be changed accordingly. The Δm^2_{23} prior was changed from $[2.30, 2.70]$ to $[2.30, 2.67]$ to decrease the maximum possible change in p-value to 5.30×10^{-4}, which is equal to the min. p-value error. This increases the required CPU time to 41% of the nominal time. The other to priors were not changed from the values in Tab 19.

B. p-value Stability

In order to assess whether the number of fake-data toys used is sufficient, the stability of the generated p-values is assessed. This is done by generating four independent sets of 20k AsimovA-constrained toys and calculating their corresponding rate-only and rate+shape p-values, following the method described in section 2, and shown in Tab 20. Qualitatively, these p-values are roughly consistent with each other as the observed variation is not large enough to change the conclusions

Set	Rate-only p-value		Rate+shape p-value	
	$\beta = 0$	$\beta = 1$	$\beta = 0$	$\beta = 1$
1	0.0364 ± 0.001	0.387 ± 0.003	0.0128 ± 0.0008	0.426 ± 0.002
2	0.0363 ± 0.001	0.386 ± 0.003	0.0134 ± 0.0008	0.420 ± 0.002
3	0.0363 ± 0.001	0.386 ± 0.003	0.0125 ± 0.0008	0.425 ± 0.002
4	0.0362 ± 0.001	0.387 ± 0.003	0.0148 ± 0.0009	0.421 ± 0.002

Table 20: p-values obtained from four separate sets of 20k AsimovA-constrained fake-data toys

made; however, as more data is collected it is possible that the variation in p-value could affect the conclusions made on the physics, for example if the p-value was close to the threshold for discovery. Therefore, it is important for future analyses to ensure that enough toys are used to keep the variation in p-value small enough that the p-values are quantitatively consistent with each other.

Additionally, the p-value is dependant on the ensemble of nuisance parameters used. As each of the four sets of fake-data toys correspond to different sets of 20k nuisance parameter throws, the p-values will not be drawn from the same underlying distribution and therefore may not be directly comparable, although the size of this effect is currently unknown. Therefore studies need to be performed during future analyses to characterise this effect.

B.1. Improved p-value Stability Study using the Full Run 1-9 Sensitivity

To assess the stability of the p-values in a more quantitative manner and with a larger set of p-values, the p-value stability study was redone during the full Run 1-9 analysis using the earlier true value of $\sin^2(\theta_{13})$ in Tab 2. Now the stability of the p-values for each test statistic and value of β are assessed by searching for outliers in sets of 10 independent p-values. Whether a p-value is considered an outlier is determined by applying a two-sided Grubbs' test for outliers, which makes use of Student's t-distribution to search for outliers in an approximately normal data set [13]:

For each test statistic and value of β, $N = 10$ independent sets of 20k AsimovA-constrained toys are generated and fluctuated (marginalised) to obtain rate-only (rate+shape) p-values, p.

Grubbs' test statistic, G is defined as

$$G = \frac{\max_{i=1,\ldots,N} |p_i - \bar{p}|}{s} \tag{18}$$

Where $\bar{p}$ is the mean p-value and s is the sample standard deviation of the p-values calculated with the tested p-value, p_i, removed.

The null hypothesis of Grubbs' test, H_0, is that none of the p-values are outliers. The alternative hypothesis, H_a, is that there is exactly one outlier. H_0 is then rejected and p_i is considered to be an outlier if the following condition is met:

$$G > \frac{N-1}{N} \sqrt{\frac{t^2_{\alpha/(2N),N-2}}{N-2+t^2_{\alpha/(2N),N-2}}} \tag{19}$$

Set	Rate-only p-value		Rate+shape p-value	
	$\beta = 0$	$\beta = 1$	$\beta = 0$	$\beta = 1$
1	0.0101 ± 0.0007	0.463 ± 0.004	0.0035 ± 0.0004	0.436 ± 0.004
2	0.0102 ± 0.0007	0.463 ± 0.004	0.0036 ± 0.0004	0.438 ± 0.004
3	0.0102 ± 0.0007	0.463 ± 0.004	0.0038 ± 0.0004	0.439 ± 0.004
4	0.0102 ± 0.0007	0.464 ± 0.004	0.0038 ± 0.0004	0.440 ± 0.004
5	0.0102 ± 0.0007	0.464 ± 0.004	0.0041 ± 0.0005	0.440 ± 0.004
6	0.0103 ± 0.0007	0.465 ± 0.004	0.0041 ± 0.0005	0.440 ± 0.004
7	0.0103 ± 0.0007	0.465 ± 0.004	0.0042 ± 0.0005	0.441 ± 0.004
8	0.0103 ± 0.0007	0.465 ± 0.004	0.0042 ± 0.0005	0.441 ± 0.004
9	0.0103 ± 0.0007	0.465 ± 0.004	0.0044 ± 0.0005	0.443 ± 0.004
10	0.0103 ± 0.0007	0.467 ± 0.004	0.0046 ± 0.0005	0.444 ± 0.004

Table 21: p-values obtained from 10 independent sets of 20k AsimovA-constrained fake-data toys produced using full Run 1-9 statistics

Analysis	β	Most Extreme p-value	$\bar{p}$	G	H_0 Rejected if $G >$	Is Outlier?
rate+shape	0	0.0046 ± 0.0005	0.0040	2.54	4.12	No
	1	0.436 ± 0.004	0.441	2.49	3.95	No
rate-only	0	0.0101 ± 0.0007	0.0103	2.42	3.95	No
	1	0.467 ± 0.004	0.464	3.25	4.12	No

Table 22: Assessment of whether the most extreme p-values from Tab 21 are considered to be outliers at a significance of $\alpha = 0.05$ using Grubbs' test for outliers

Where $t^2_{\alpha/(2N),N-2}$ is the critical value of Student's t-distribution with $N-2$ d.o.f and significance level α.

The p-values for each of the 10 toy sets, calculated for both rate-only and rate+shape analyses under both the $\beta = 0$ and $\beta = 1$ hypotheses, are shown in Tab 21. The most extreme p-values from this table are assessed in Tab 22 and it is shown that, at a significance level of $\alpha = 0.05$, none of the p-values are considered to be outliers. Therefore 20k fake data toy sets is considered to be sufficient at this significance level.

C. Toy Generation Method Comparison

During the Run 1-9c analysis the Asimov and data p-values (see sections 3 and 4, respectively) were compared to those produced by the P-theta group and differences of $< 0.1\,\sigma$ were found. It was suggested that these differences may have been due to the different toy generation methods used by VALOR and P-theta. The equivalence of the two methods is assessed in this section.

In this analysis, toys are generated according to the procedure detailed in section 2, where toys are accepted/rejected based on the likelihood of the control sample data given the toy. p-values are then calculated using only the accepted toys. This method will be referred to as 'MC sampling'. The analysis performed by the P-theta group does not use this accept/reject method, instead all toys are accepted the likelihood of the control sample data given the toy is stored and used as a weight when calculating p-values. This method will be referred to as 're-weighting'.

The p-values produced using these two methods will be directly compared using a single set of 3.5M AsimovA-constrained toys. To generate toys, steps 1-4 of the method from section 2 are used, and all 3.5M toys are accepted. In addition, the control sample likelihood of the Asimov A dataset given the toy is stored as the weight, w, for each toy.

After generating the toys, they are all fluctuated (marginalised) to calculate their rate-only (rate+shape) test statistics. Then, without discarding any toys, each toy is labelled as accepted or rejected using a probability of $\frac{w}{\lambda_{max}}$. Two different p-values are then calculated:

- MC sampling p-value using only accepted toys e.g. $p(\text{right-tail}) = \frac{N^{\text{R}}_{\text{acc.}}}{N^{\text{R}}_{\text{acc.}}+N^{\text{L}}_{\text{acc.}}}$ where $N^{\text{R}}_{\text{acc.}}$ denotes the number of accepted toys in the right tail
- Re-weighting p-value using all toys e.g. $p(\text{right-tail}) = \frac{\sum_{i=1}^{N^{\text{R}}_{\text{all}}} w_i}{\sum_{i=1}^{N^{\text{R}}_{\text{all}}} w_i + \sum_{i=1}^{N^{\text{L}}_{\text{all}}} w_i}$

By comparing the MC sampling p-value to the re-weighting p-value for each analysis and hypothesis, the assumption that the two generation methods are equivalent is tested.

The test statistic distributions of sampled and re-weighted toys are shown for the rate-only analysis in Fig 12 and for the rate+shape analysis in Fig 13. The test statistic distributions are in good agreement, with small differences from the low statistics of the set of accepted toys: Out of the 3.5M toys generated, the rate-only analysis accepted 8675 and 9005 toys for $\beta = 0$ and $\beta = 1$, respectively, while the rate+shape analysis accepted 8669 and 9005 toys for $\beta = 0$ and $\beta = 1$, respectively.

The resulting p-values are shown in Tab 23. As all of the re-weighting p-values are within the error range of the MC sampling p-values, the two toy generation methods are considered to be equivalent.

Analysis	Fake data β	Toy Gen. Method	p-value	σ
rate-only	0	MC sampling	0.010 ± 0.001	2.56 ± 0.04
		re-weighting	0.01028 ± 0.00005	2.566 ± 0.002
	1	MC sampling	0.465 ± 0.005	0.730 ± 0.009
		re-weighting	0.4645 ± 0.0003	0.7315 ± 0.0004
rate+shape	0	MC sampling	0.0040 ± 0.0007	$2.87^{+0.06}_{-0.05}$
		re-weighting	0.0038 ± 0.0003	2.90 ± 0.02
	1	MC sampling	0.436 ± 0.005	0.78 ± 0.009
		re-weighting	0.438 ± 0.002	$0.778^{+0.004}_{-0.003}$

Table 23: Comparison of the p-values obtained using two different toy generation methods, MC sampling and re-weighting.

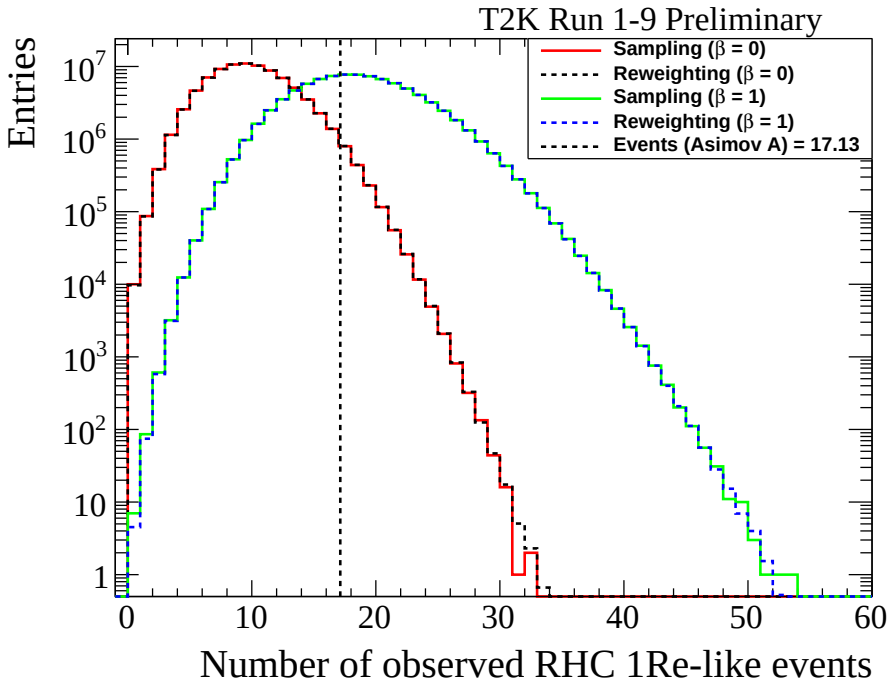


Figure 12: Comparison of rate-only test statistic distributions for toy experiments generated by MC sampling and re-weighting. The distribution of the total number of events in the antineutrino mode single-ring e-like sample for both the $\beta = 0$ and $\beta = 1$ hypotheses are produced using full Run 1-9 statistics. The dashed line corresponds to the total number of events given by the Asimov A data set.

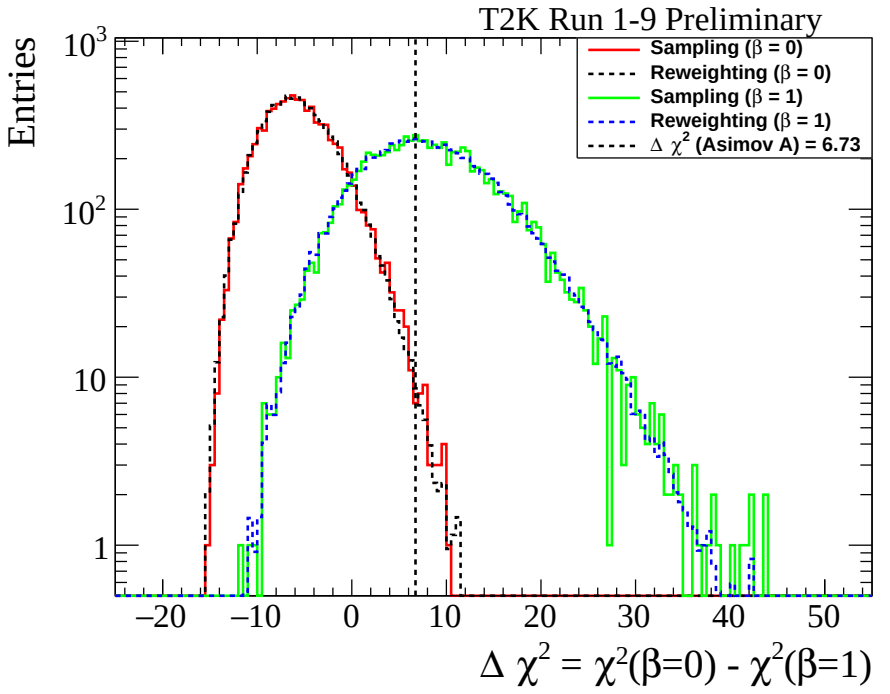


Figure 13: Comparison of rate+shape test statistic distributions for toy experiments generated by MC sampling and re-weighting. The distribution of $\Delta\chi^2$ values for both the $\beta = 0$ and $\beta = 1$ hypotheses are produced using full Run 1-9 statistics. The dashed line corresponds to the $\Delta\chi^2$ value given by the Asimov A data set.

D. Comparison of Full Run 1-9 vs. Run 1-9c Results

D.1. Comparison of Expected Sensitivities

Figs 14 and 15 show the expected Asimov A sensitivities compared between Run 1-9d and Run 1-9c for the rate-only and rate+shape analyses, respectively. For both analyses, the distributions have broadened and their separation has increased. A comparison of the p-values and test statistics is shown in Tab 24 for the rate-only analysis and for Tab 25 for the rate+shape analysis.

Compared to the Run 1-9c analysis, the Run 1-9d analysis expects a stronger exclusion of the $\beta = 0$ hypothesis in both analyses. This is driven by the combination of two factors:

1. The increased statistics further separates the two distributions, as for $\beta = 0$ it contains only background events while for $\beta = 1$ it contains both signal and background events.
2. The Asimov dataset assumes the $\beta = 1$ hypothesis to be true and uses most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [6], so its test statistic value is close to the mean value of the $\beta = 1$ distribution for each run period.

Both the Run 1-9d and Run 1-9c analyses expect to disfavour the $\beta = 1$ hypothesis at an approximately equal level due to the second point above.

D.2. Comparison of Data Results

Figs 16 and 17 show the observed distributions and test statistics compared between Run 1-9d and Run 1-9c for the rate-only and rate+shape analyses, respectively. A comparison of the p-values and test statistics is shown in Tab 26 for the rate-only analysis and for Tab 27 for the rate+shape analysis.

For both rate-only and rate+shape analyses, the $\beta = 0$ ($\beta = 1$) hypothesis is excluded at a greater (lesser) significance in the Run 1-9d analysis than in Run 1-9c. There are two reasons for this change. The first reason is shared by both rate-only and rate+shape analyses - the increased statistics further separates the two distributions (as explained in sec D.1).

For the rate-only analysis the second reason is that the event rate has increased from $8.01 \frac{\text{events}}{10^{21}\text{POT}}$ in Run 1-9c to $9.18 \frac{\text{events}}{10^{21}\text{POT}}$ in Run 1-9d, shifting the rate-only test statistic closer to the mean value of the $\beta = 1$ distribution and further from the mean of the $\beta = 0$ distribution.

For the rate+shape analysis, the second reason for this change can be seen by comparing Fig 11d to Fig 3d. It can be seen that of the six newly collected (Run 9d) events in the antineutrino mode single-ring electron-like sample, five events are in the most likely region as predicted by the Asimov A dataset under the assumption of the $\beta = 1$ hypothesis. This is in contrast to the Run 1-9c data, where most of the nine events lay outside this region. This has the effect of changing the data $\Delta\chi^2$ value from the Run 1-9c value of -1.502 to the Run 1-9d value of 3.811. This $\Delta\chi^2$ value is now close to the mean of the $\beta = 1$ distribution, rather than almost midway between the two distributions, as was the case in Run 1-9c. This change is further demonstrated by the Bayes factor of 0.149 which corresponds to a substantial preference for the $\beta = 1$ hypothesis over $\beta = 0$, while for the Run 1-9c analysis the Bayes factor was 2.12 meaning a poor preference for the $\beta = 0$ hypothesis over $\beta = 1$.

Run	RHC POT	N_{exp}	$\beta = 0$		$\beta = 1$	
			p-value	σ	p-value	σ
1-9d	1.6346×10^{21}	16.8	0.0185 ± 0.0009	2.36 ± 0.02	0.379 ± 0.003	0.88 ± 0.01
1-9c	1.1236×10^{21}	11.8	0.036 ± 0.001	2.07 ± 0.01	0.386 ± 0.003	0.87 ± 0.01
1-7	7.471×10^{20}	6.008	0.0511 ± 0.00002	1.9506 ± 0.0002	0.5022 ± 0.00005	0.67103 ± 0.0008

Table 24: Expected rate-only p-values and the number of events from the full Run 1-9 data set compared to the Run 1-9c and Run 1-7 data set used for previous iterations of this analysis, in Summer 2018 and 2016[1], respectively.

Run	RHC POT	$\Delta\chi^2_{\text{exp}}$	$\beta = 0$		$\beta = 1$		B_{01}
			p-value	σ	p-value	σ	
1-9d	1.6346×10^{21}	6.27	0.0057 ± 0.0005	2.76 ± 0.03	0.409 ± 0.004	0.83 ± 0.01	0.0436
1-9c	1.1236×10^{21}	4.592	0.0128 ± 0.0008	2.49 ± 0.02	0.426 ± 0.003	0.80 ± 0.01	0.101
1-7	7.471×10^{20}	2.610	0.043 ± 0.002	2.02 ± 0.02	0.444 ± 0.005	0.765 ± 0.008	0.271

Table 25: Expected rate+shape p-values, $\Delta\chi^2$ and Bayes factor from the full Run 1-9 data set compared to the Run 1-9c and Run 1-7 data set used for previous iterations of this analysis, in Summer 2018 and 2016[1], respectively.

Run	RHC POT	N_{obs}	$\beta = 0$		$\beta = 1$	
			p-value	σ	p-value	σ
1-9d	1.6346×10^{21}	15	0.059 ± 0.002	$1.89^{+0.02}_{-0.01}$	0.321 ± 0.003	0.99 ± 0.01
1-9c	1.1236×10^{21}	9	0.209 ± 0.003	1.26 ± 0.01	0.210 ± 0.003	1.25 ± 0.01
1-7	7.471×10^{20}	4	0.4100 ± 0.00005	0.8239 ± 0.00009	0.2155 ± 0.00004	1.2486 ± 0.0001

Table 26: Observed rate-only p-values and number of events from the full Run 1-9 data set compared to the Run 1-9c and Run 1-7 data set used for previous iterations of this analysis, in Summer 2018 and 2016[1], respectively.

Run	RHC POT	$\Delta\chi^2_{\text{obs}}$	$\beta = 0$		$\beta = 1$		B_{01}
			p-value	σ	p-value	σ	
1-9d	1.6346×10^{21}	3.811	0.0163 ± 0.0009	2.40 ± 0.02	0.300 ± 0.004	1.04 ± 0.01	0.149
1-9c	1.1236×10^{21}	-1.502	0.215 ± 0.003	1.24 ± 0.001	0.092 ± 0.003	1.68 ± 0.02	2.12
1-7	7.471×10^{20}	-1.930	0.3742 ± 0.0048	0.888 ± 0.009	0.099 ± 0.003	1.65 ± 0.01	2.62

Table 27: Observed rate+shape p-values, $\Delta\chi^2$ and Bayes factor from the full Run 1-9 data set compared to the Run 1-9c and Run 1-7 data set used for previous iterations of this analysis, in Summer 2018 and 2016[1], respectively.

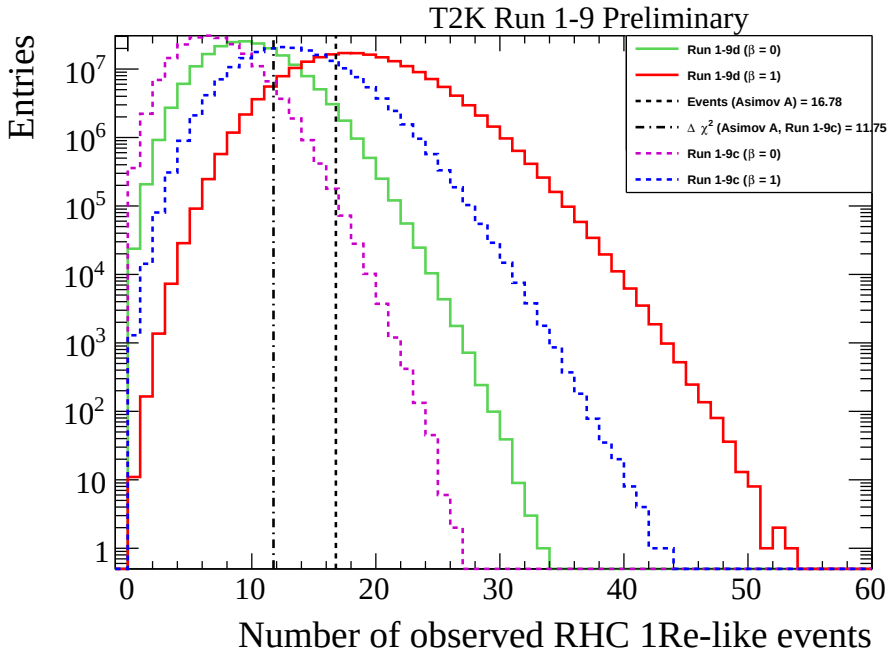


Figure 14: Comparison between the expected Run 1-9d and Run 1-9c distributions of the total number of events in the antineutrino mode single-ring e-like sample. The distributions were produced with 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses using the method outlined in section 2.1 and the Asimov A spectra to calculate the 4-sample likelihood.

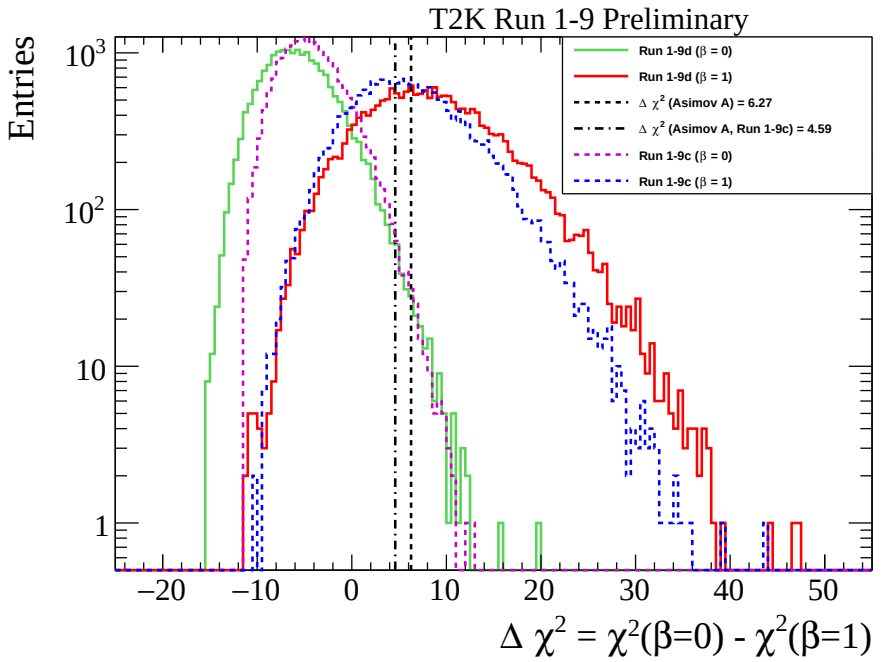


Figure 15: Comparison between the expected Run 1-9d and Run 1-9c distributions of Asimov-constrained $\Delta\chi^2$ values. The distributions were produced with 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses using the method outlined in section 2.2 with the Asimov A spectra to calculate the 4-sample likelihood.

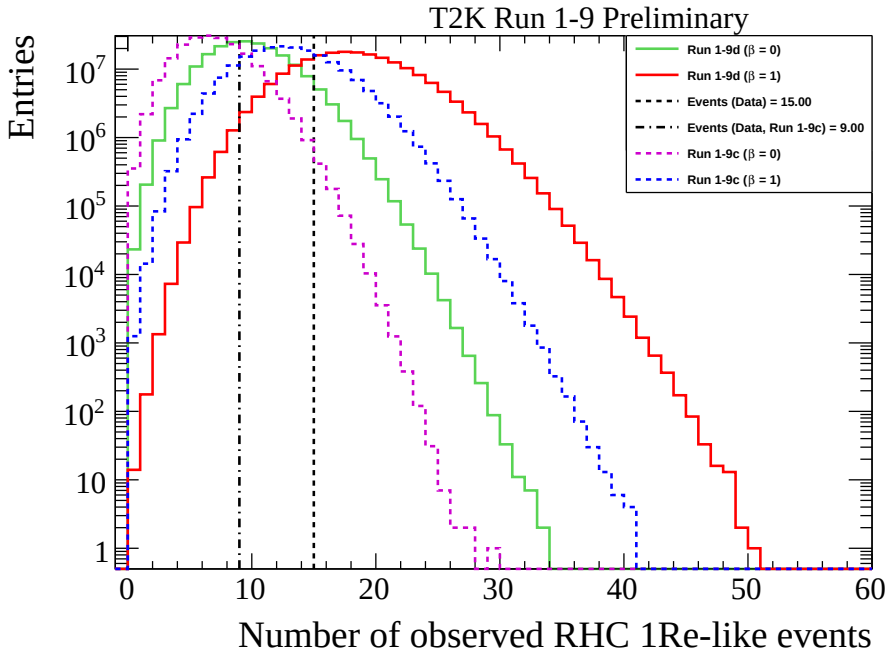


Figure 16: Comparison between the observed Run 1-9d and Run 1-9c distributions of the total number of events in the antineutrino mode single-ring e-like sample. The distributions were produced with 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses using the method outlined in section 2.1 and the data spectra to calculate the 4-sample likelihood.

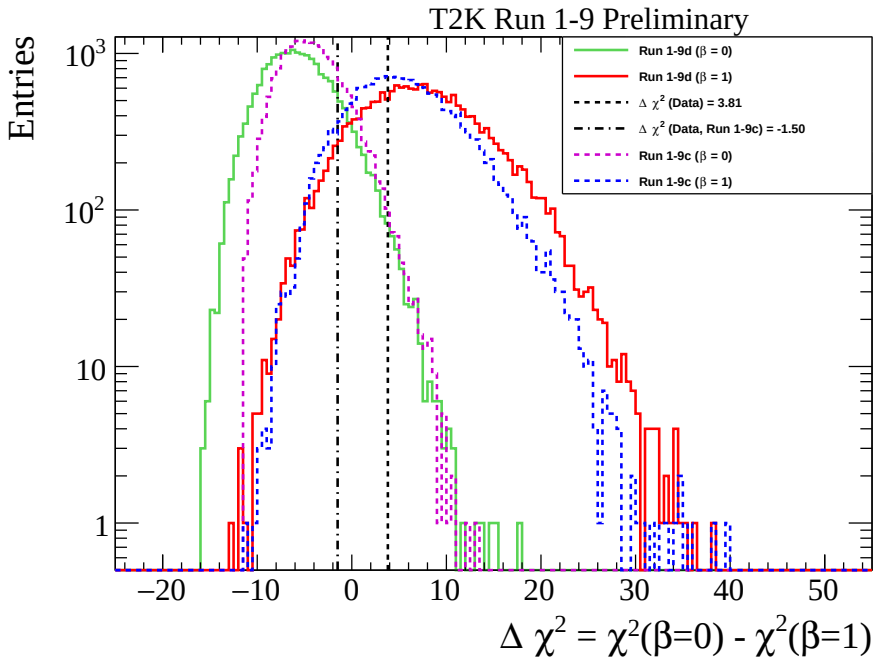


Figure 17: Comparison between the observed Run 1-9d and Run 1-9c distributions of data-constrained $\Delta\chi^2$ values. The distributions were produced with 20k toy experiments for both the $\beta = 0$ and $\beta = 1$ hypotheses using the method outlined in section 2.2 with the data spectra to calculate the 4-sample likelihood.

E. Comparison Between VALOR and P-theta Results using Full Run 1-9 Statistics

To validate the results of this analysis, they are compared to the results obtained by the P-theta group. The P-theta analysis [8] uses electron-like spectra binned in (p, θ) variables, while the VALOR analysis uses (E_{rec}, θ). The use of different binning variables will cause the calculation of the control sample likelihood weights (in both rate-only and rate+shape analyses) and marginal likelihood to be different between VALOR and P-theta. Therefore the p-values in the P-theta tech note (see ref [8]) are not directly comparable to the p-values in this document. Instead the P-theta group has duplicated their rate+shape analyses using (E_{rec}, θ) binning to produce the p-values which are considered in this section.

E.1. Comparison of Expected Sensitivities

The P-theta analysis was completed before the Asimov A true value of $\sin^2 2\theta_{13}$ was updated to 0.0830, so used the previous value of 0.0856. In order to have directly comparable p-values, the VALOR p-values shown in this section were produced using the same value of $\sin^2 2\theta_{13}$ as used in the P-theta analysis, so are not the same p-values shown in section 6.1.

For the rate-only analysis, VALOR expects a test statistic of 17.13 events, while P-theta expects 17.15 events. For the rate+shape analysis, VALOR expects a test statistic of $\Delta\chi^2 = 6.73$, while P-theta expects $\Delta\chi^2 = 6.69$. Figs 18 and 19 show the expected rate-only and rate+shape test statistic distributions. As Tab 28 shows, the p-values produced by VALOR and P-theta are in good agreement.

Analysis	Fake data β	Group	p-value	σ
rate-only	0	VALOR	0.0102 ± 0.0007	2.57 ± 0.02
		P-theta	0.0109	2.55
	1	VALOR	0.465 ± 0.004	0.73 ± 0.01
		P-theta	0.453	0.75
rate+shape	0	VALOR	0.0046 ± 0.0005	$2.83^{+0.04}_{-0.03}$
		P-theta	0.00369	2.90
	1	VALOR	0.439 ± 0.004	0.77 ± 0.01
		P-theta	0.429	0.79

Table 28: Comparison of the p-values given by the Asimov A sensitivity studies from the VALOR and P-theta groups.

E.2. Comparison of Data Results

For the rate-only analysis the observed test statistic, the event rate in the antineutrino mode single-ring e-like sample (15 events) is the same for VALOR and P-theta, as expected. For the rate+shape analysis, differences in the ensemble of marginalisation toys causes slightly different observed $\Delta\chi^2$ values, 3.81 for VALOR and 4.09 for P-theta. Figs 20 and 21 show the observed rate-only and rate+shape test statistic distributions. As Tab 29 shows, the p-values produced by VALOR and P-theta are in good agreement, with the exception of the rate-only $\beta = 1$ p-value. This discrepancy is thought to be due to the fact that for the rate-only analysis, the P-theta group has only used (p, θ) binning to calculate the control sample likelihood weights (see appendix C),

and an equivalent rate-only analysis using (E_{rec}, θ) binning is not available. This difference is not seen in the rate-only sensitivity study, similarly to the Run 1-9 joint analysis [3], where differences in the oscillation parameter confidence intervals produced by VALOR and P-theta are seen in the data fits but not in the sensitivity study.

Analysis	Fake data β	Group	p-value	σ
rate-only	0	VALOR	0.059 ± 0.002	$1.89^{+0.02}_{-0.01}$
		P-theta	0.0686	1.82
	1	VALOR	0.321 ± 0.003	0.99 ± 0.01
		P-theta	0.246	1.16
rate+shape	0	VALOR	0.0163 ± 0.0009	2.40 ± 0.02
		P-theta	0.0155	2.42
	1	VALOR	0.300 ± 0.004	1.04 ± 0.01
		P-theta	0.300	1.04

Table 29: Comparison of the observed p-values from the VALOR and P-theta groups.

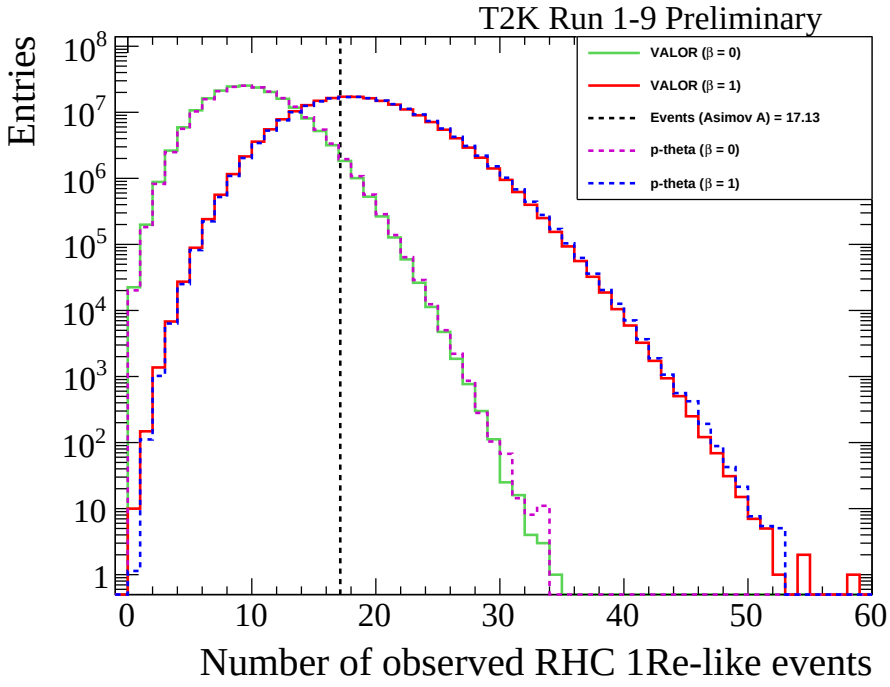


Figure 18: Comparison between the expected VALOR and P-theta distributions of the total number of events in the antineutrino mode single-ring e-like sample. The VALOR distributions were produced with 20k toy experiments using MC sampling (see appendix C). The P-theta distributions were produced with 100k toy experiments using re-weighting. Asimov A spectra were used to calculate the 4-sample likelihood.

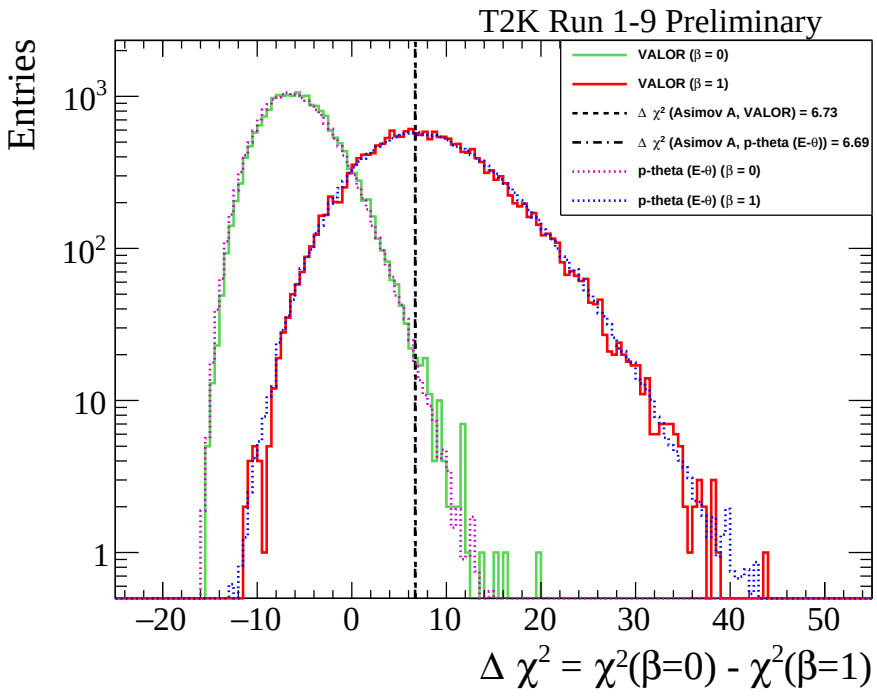


Figure 19: Comparison between the expected VALOR and P-theta $\Delta\chi^2$ distributions. The VALOR distributions were produced with 20k toy experiments using MC sampling (see appendix C). The P-theta distributions were produced with 2000k toy experiments using re-weighting. Asimov A spectra were used to calculate the 4-sample likelihood.

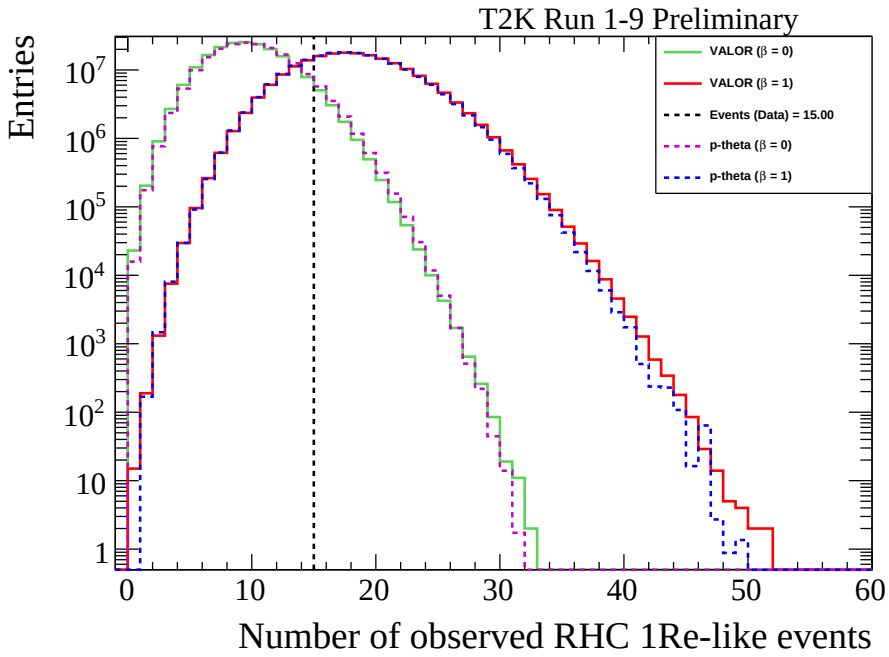


Figure 20: Comparison between the observed VALOR and P-theta distributions of the total number of events in the antineutrino mode single-ring e-like sample. The VALOR distributions were produced with 20k toy experiments using MC sampling (see appendix C). The P-theta distributions were produced with 100k toy experiments using re-weighting. Observed Run 1-9 spectra were used to calculate the 4-sample likelihood.

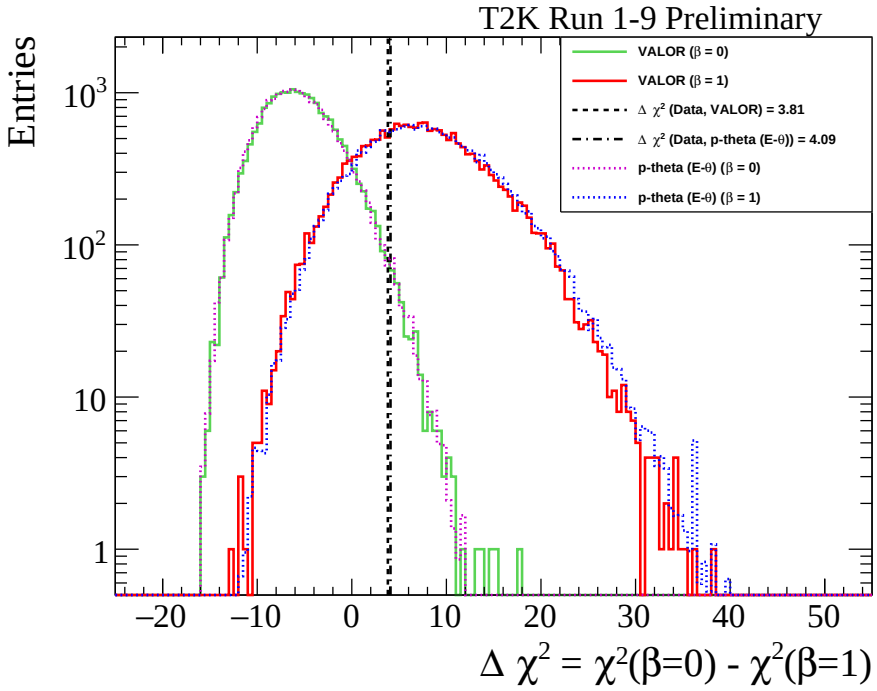


Figure 21: Comparison between the observed VALOR and P-theta $\Delta\chi^2$ distributions. The VALOR distributions were produced with 20k toy experiments using MC sampling (see appendix C). The P-theta distributions were produced with 2000k toy experiments using re-weighting. Observed Run 1-9 spectra were used to calculate the 4-sample likelihood.